

Quantum Fisher Information as a Thermal Probe in Frustrated Magnets through Insights from Quantum Spin Ice

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Quantum Fisher information (QFI) is a measure of multipartite entanglement accessible via inelastic neutron scattering. Here we demonstrate that QFI reveals thermal and dynamical properties of quantum spin ice (QSI), a three-dimensional quantum spin liquid with fractionalized excitations. By developing a multi-directed loop update quantum Monte Carlo algorithm, along with exact diagonalization and gauge mean-field theory, we compute the QFI for the pyrochlore lattice. The temperature and momentum dependence of QFI maps the phase diagram, distinguishing the ferromagnetic ordered phase, its critical region, the zero-flux QSI, and the π -flux QSI. QFI also captures two crossover scales: from trivial paramagnet to classical spin ice, then to QSI. We discuss the π -flux QSI in light of experiments on cerium-based pyrochlores. Our results suggest that QFI not only detects entanglement but also serves as a sensitive thermal and dynamical probe for frustrated quantum magnets.

INTRODUCTION

Entanglement is arguably the most important concept in quantum physics. It is at the heart of our modern understanding of phases of matter and transitions between them. As a notable example, entanglement is the fundamental characteristic of quantum spin liquids (QSLs) [1–4] — paramagnetic phases of frustrated spin systems that fail to magnetically order down to zero temperature, and host fractional excitations and emergent gauge fields [5–9]. QSLs are characterized by long-range entanglement, which may be measured for gapped QSL by a non-vanishing topological entanglement entropy [4, 10–13]. However, despite its role as a theoretical cornerstone and its usefulness in numerical studies [12–17], topological entanglement entropy is not experimentally accessible in solid-state platforms, where only local correlations are typically measured.

One still has experimental access to other measures of entanglement. A particularly useful one that has recently been at the forefront of the experimental search for QSLs is the quantum Fisher information (QFI) [18–24]. Initially employed to define the maximal achievable precision in parameter estimation for a given quantum state in the quantum metrology community [25–30], the QFI is directly related to the dynamical susceptibility [31], which is routinely measured in inelastic neutron scattering experiments [32–34]. The QFI density f_Q provides a lower bound on the multipartite entanglement in the system [35, 36]. This bound can be used to experimentally differentiate between QSLs and other, more trivial states, such as random singlet states driven by strong disorder [24, 37]. These states may otherwise be challenging to distinguish experimentally, as they can both lead to similar signatures, such as continua of excitations in inelastic neutron scattering or a lack of experimentally observable finite-temperature phase transitions [38–44]. Nevertheless, it should be emphasized that a large value of the QFI, although promis-

ing, does not provide evidence for the realization of a QSL in and of itself. Indeed, trivial ordered states sufficiently close to a quantum critical point may have an arbitrarily large value of f_Q [31]. Be as it may, if precise theoretical predictions for the momentum and temperature dependence of the QFI for prospective QSLs exist, they offer stringent quantitative predictions which, if measured, may provide significantly more convincing evidence than qualitative features, like the presence of broad continua of excitations.

It is worth noting that the QFI is not meant to be a universal entanglement classifier; rather, it provides an operator-resolved and experimentally aligned witness for multipartite entanglement depth. Compared with other witnesses, QFI is distinguished in collective spin systems because it is simultaneously (i) well-defined for mixed thermal states, (ii) scalable to large many-body settings without requiring reduced density matrices, and (iii) directly linked to the excitation spectrum through a frequency integral of the dynamical response. By contrast, pairwise measures such as concurrence [37, 45] probe only two-spin entanglement, they neither certify multipartite entanglement depth nor remain informative when entanglement is predominantly multipartite and delocalized, as is typical in collective correlated phases.

In this letter, we make such detailed predictions for the QFI on one of the most paradigmatic QSL: quantum spin ice (QSI) [46–51]. QSI is a three-dimensional QSL that is the ground state of an XXZ model with dominant Ising and subleading transverse couplings on the pyrochlore lattice. It realizes the deconfined (Coulomb) phase of compact $U(1)$ gauge theory and, as such, hosts emergent photon excitations, spin-1/2 spinons that act as emergent electric charges, and magnetic monopoles [52]. QSI is the ideal platform for making specific predictions for the QFI, as it is one of the few known examples of an experimentally relevant model that stabilizes a well-understood QSL that is numerically accessible over a parameter regime (i.e., ferromagnetic

transverse couplings) with sign-problem-free quantum Monte Carlo (QMC) [52–57]. Furthermore, several compounds have historically been considered as possible experimental realization of QSI such as $\text{Tb}_2\text{Ti}_2\text{O}_7$ [58–62], $\text{Pr}_2(\text{Sn},\text{Zr},\text{Hf})_2\text{O}_7$ [63–68], and $\text{Yb}_2\text{Ti}_2\text{O}_7$ [50, 69–74]. The most recently considered candidate materials are the Cerium-based pyrochlore compounds $\text{Ce}_2(\text{Zr},\text{Hf},\text{Sn})_2\text{O}_7$ [75–85]. For this last family of compounds, no magnetic order has been reported in $\text{Ce}_2(\text{Zr},\text{Hf})_2\text{O}_7$. Experimental determinations of their microscopic couplings indicate that $\text{Ce}_2\text{Zr}_2\text{O}_7$ and $\text{Ce}_2\text{Hf}_2\text{O}_7$ likely fall in a region of parameter space that stabilizes, so-called, π -flux QSI [78, 83–88] where a constant flux of the emergent gauge field is threading hexagonal plaquettes such that translation acts projectively on spinons excitations [89–95]. Energy integrated and inelastic neutron scattering measurements consistent with theoretical predictions have also been reported [78, 80, 84, 87, 95, 96], and a cubic scaling of the low-temperature heat capacity has even been recently measured in $\text{Ce}_2\text{Zr}_2\text{O}_7$ [80]. Precise theoretical predictions for the QFI should thus be experimentally measurable and provide rigorous tests for these and future candidate materials.

We employ large-scale QMC simulations [97–100] to evaluate the QFI of the XXZ model with ferromagnetic transverse coupling. To this end, we develop a multi-directed loop (MDL) update algorithm to efficiently sample the highly frustrated (3+1)d configurational space; the details are provided in the Supplemental Material (SM) [101]. Exact diagonalization (ED) is also used to compute the QFI [102, 103] for both anti-ferromagnetic and ferromagnetic transverse couplings. We show that ED and QMC results are consistent for ferromagnetic XY exchange, which justifies extending our ED result to the anti-ferromagnetic exchange case, where π -flux QSI is the ground state. These numerical results are further compared with predictions from the gauge mean-field theory (GMFT) parton construction [89, 90, 94, 95, 104–107], which provides a mapping from the spin Hamiltonian to a lattice U(1) gauge theory coupled to bosonic matter fields in the low-temperature limit.

We find that the QFI at finite temperature is sensitive to thermal phase transitions and crossovers, as well as regions of large critical fluctuations, depending on the momentum position. For example, the QFI associated with the transverse spin components $S^\pm$ at the $\Gamma = (0, 0, 0)$ point clearly maps out the thermal phase diagram of the pyrochlore XXZ model, revealing the two crossover temperature scales between the high-temperature paramagnet and classical spin ice as well as between classical spin ice and QSI [52, 56]. Its evolution also delineates the ferromagnetic (FM) thermal phase transition for large ferromagnetic transverse couplings [53, 55]. In contrast, the QFI at the $\Gamma' = (4\pi, 4\pi, 0)$ point appears to be sensitive to critical fluctuations near the phase transition between QSI and the ferromagnetic ordered phases. Altogether, our work provides the first large-scale, unbiased computation of experimentally accessible entanglement properties of QSI, offering exhaustive numerical results that can be quantitatively compared with future experimental results on current candidate

materials.

RESULTS

Model

The pyrochlore lattice is a face-centered cubic lattice with four sublattices per unit cell, forming a network of corner-sharing tetrahedra. We consider spin-1/2 on the pyrochlore lattice that are interacting through an XXZ model of the form

$$H = J_z \sum_{\langle i,j \rangle} S_i^z S_j^z - J_\pm \sum_{\langle i,j \rangle} (S_i^+ S_j^- + S_i^- S_j^+), \quad (1)$$

where the summation $\langle i, j \rangle$ runs over all the nearest-neighbor pairs. We focus on the regime with a dominant antiferromagnetic Ising coupling (i.e., $J_z > |J_\pm| > 0$), which provides geometric frustration within each tetrahedron and energetically favors “classical spin ice configurations” where $\sum_{i \in \text{tet}} S_i^z = 0$ with the sum over all spins in a given tetrahedron. In the perturbative regime $|J_\pm| \ll J_z$, the transverse coupling allows for tunneling between the classically degenerate configurations that respect this local energetic constraint [46]. This tunnelling stabilizes 0-flux QSI (QSI_0) for $0 < J_\pm/J_z < 0.052$ at low temperatures ($T \lesssim 12|J_\pm^3/J_z^2|$) as shown in QMC [52–56]. When $-1 < J_\pm/J_z < 0$, π -flux QSI (QSI_π) is instead stabilized, where the flux refers to the emergent static magnetic flux threading the hexagonal plaquettes of the pyrochlore lattice in the ground state [46, 89, 92, 94, 95, 108–110]. For $J_\pm/J_z > 0.052$, the system undergoes a fluctuation-induced first-order transition [111, 112] into an XY ferromagnetic (FM) ordered phase below a critical temperature T_c [53, 55, 56]. Hereafter, we set $J_z = 1$ as the unit of energy. For reference, the dominant exchange is usually slightly smaller than 1 K in Cerium-based dipolar-octupolar candidates [78, 83–88].

To quantify the entanglement properties of a given state, the QFI provides a lower bound on multipartite entanglement, also known as entanglement depth [37, 45, 113]. For a collective generator $\mathcal{O} = \sum_{i=1}^N \mathcal{O}_i$ with local eigenvalue range $\Delta\lambda = \lambda_{\max} - \lambda_{\min}$, any m -producible state ρ obeys [20, 35, 36, 45, 114]

$$\frac{F_Q[\rho, \mathcal{O}]}{N} \leq m(\Delta\lambda)^2. \quad (2)$$

Equivalently, with $f_Q(\mathcal{O}) := F_Q[\rho, \mathcal{O}]/N$, we can define the normalized QFI (nQFI) as the lower bound of entanglement depth, such that

$$\text{nQFI}(\mathcal{O}) := \frac{f_Q(\mathcal{O})}{(\Delta\lambda)^2} > m \implies \text{entanglement depth} \geq m + 1, \quad (3)$$

For the operator $\mathcal{O} = S_{\mathbf{q}}^\alpha := \sum_i S_{\mathbf{R}_i}^\alpha e^{i\mathbf{q}\cdot\mathbf{R}_i}$, the QFI density is related to the dynamical structure factor $A^\alpha(\mathbf{q}, \omega) := \frac{1}{2\pi N} \int dt \langle S_{\mathbf{q}}^{\alpha\dagger}(t) S_{\mathbf{q}}^\alpha(0) \rangle e^{i\omega t}$ at momentum $\mathbf{q}$ by [31]

$$f_Q(S_{\mathbf{q}}^\alpha, T) = 4 \int_0^\infty d\omega \tanh\left(\frac{\omega}{2T}\right) (1 - e^{-\omega/T}) A^\alpha(\mathbf{q}, \omega). \quad (4)$$

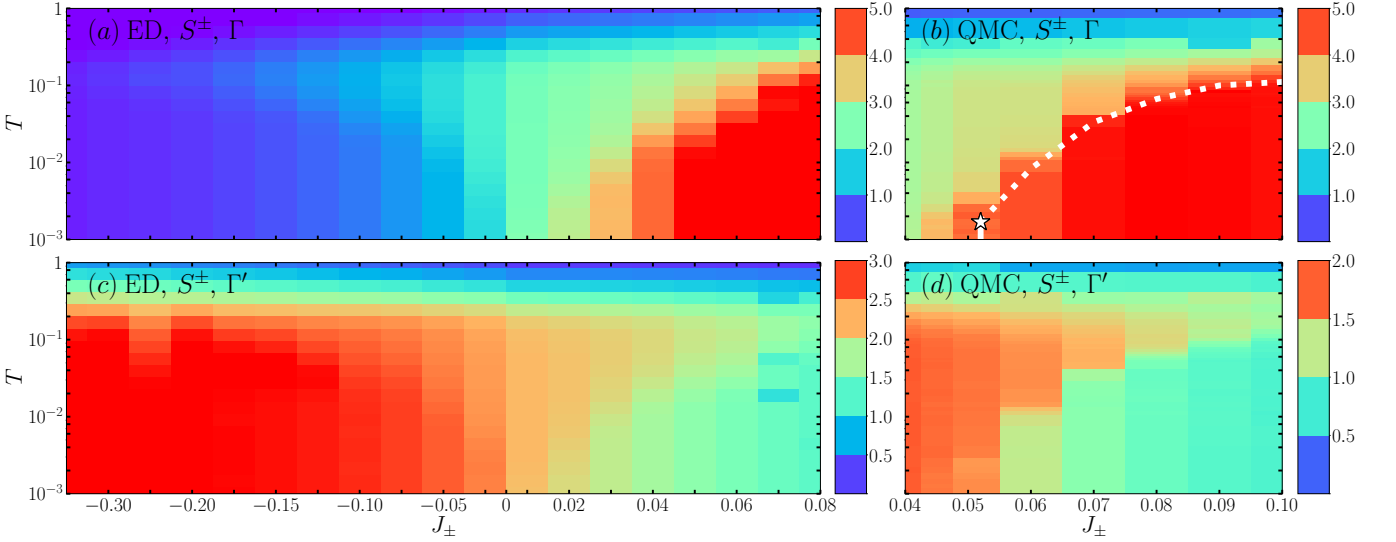


FIG. 1. **Heat maps of the QFI as functions of temperature T and $J_{\pm}$.** Panels (a) and (b) show the QFI density $f_Q(S_{\mathbf{q}}^{\pm}, T)$ in the $S^{\pm}$ channel at $\Gamma = (0, 0, 0)$, while panels (c,d) show it at $\Gamma' = (4\pi, 4\pi, 0)$. Panel (a) and (c) are obtained from the ED calculation of 16-site cluster with $J_{\pm}$ ranging from -0.045 to 0.08 , and panels (b) and (d) are from the QMC simulation of $4 \times L^3$ sizes ($L = 4$) with $J_{\pm}$ ranging from 0.04 to 0.10 . In panels (a) and (b), $f_Q(S_{\Gamma}^{\pm}, T)$ maps out the thermodynamic phase boundaries between the ferromagnetic phase (FM) and QSI_0 . The temperature dependence of $f_Q(S_{\Gamma}^{\pm}, T)$ from QMC further discerns the crossover temperature scales between the high-temperature paramagnetic regime to classical spin ice and eventually to the QSI_0 regime (see also Fig. 2 (a)). In panels (c) and (d), $f_Q(S_{\Gamma'}^{\pm}, T)$ reflects the strength of the fluctuations in the thermal and quantum phase diagram, with strong QFI at the classical critical region above the FM phase and stronger QFI in the classical spin ice and QSI_0 region. Moreover, the strongest QFI signal, represented by the red region in panel (c) for $J_{\pm} < 0$, reflects the QSI_{π} regime (see also Fig. 2 (b)). In panel (b), the white dashed line denotes the phase boundary between the paramagnetic and FM phases, whereas the white solid line ($J_{\pm,c} = 0.052$) represents the transition between the FM and QSI_0 phases. The white star marks the crossover temperature scale from the CSI regime to the QSI_0 regime ($T = 12 \times 0.052^3 \approx 0.00169$) at $J_{\pm,c}$.

Here, T is the temperature and α labels different pseudospin components, such as $\alpha \in \{x, y, z\}$. We see that the QFI density $f_Q(S_{\mathbf{q}}^{\alpha}, T)$ can be obtained by integrating $A^{\alpha}(\mathbf{q}, \omega)$ over all frequencies at a fixed temperature and momentum. Experimental measurements of the QFI have already been reported for quasi-1D KCuF_2 [20] and for 2D frustrated triangular lattice material KYbSe_2 [115].

To simulate the QSI_0 phase, we developed a MDL-QMC update that enables efficient simulations of the QSI_0 phase on the pyrochlore lattice. This task is difficult to achieve with the conventional directed loop update algorithm as in the QSI_0 phase, the simplest quantum fluctuation that connects different classical spin-ice states involves six spin flips around a hexagon, which is hard to realize by a simple directed loop update. Our MDL algorithm overcomes this limitation by allowing the insertion of multiple operator pairs, thereby naturally generating higher-order processes during the Monte Carlo update and making the QSI_0 phase accessible (see SM [101]). We then measure the imaginary time correlation function $G^{\pm} = \frac{1}{2N} \sum_{\gamma, \nu} \langle S_{-\mathbf{q}, \gamma}^{\pm}(\tau) S_{\mathbf{q}, \nu}^{\mp}(0) + S_{-\mathbf{q}, \gamma}^{\mp}(\tau) S_{\mathbf{q}, \nu}^{\pm}(0) \rangle$ and $G^z = \frac{1}{N} \sum_{\gamma, \nu} \langle S_{-\mathbf{q}, \gamma}^z(\tau) S_{\mathbf{q}, \nu}^z(0) \rangle$, where γ, ν label the four pyrochlore sublattices, the momentum-transfer $\mathbf{q}$ is measured in the 3D pyrochlore Brillouin zone (BZ), and $\tau \in [0, \beta]$ denotes the imaginary time with $\beta = 1/T$ the inverse temperature. We utilize the stochastic analytic continuation (SAC) scheme [116–120] to convert the imaginary time correlation

function to a real-frequency dynamic structure factor. This QMC+SAC scheme has been successfully applied to a variety of lattice models, producing reliable spectral properties ranging from magnon and amplitude modes in a magnetically ordered state [121, 122] to fractionalized excitations in QSL and QSI models [52, 123–125].

We further compute the QFI density using ED and GMFT. The ED calculations are performed over a 16-site periodic cubic conventional cluster (see SM [101]), providing access to momentum positions Γ, Γ' , and X . We first compare the ED and GMFT results with our QMC data before extending these calculations into the π -flux $J_{\pm} < 0$ regime, where most QSI candidate materials reside and QMC encounters a sign problem.

Numerical Results

Our main results are summarized in Fig. 1. The white dashed line marks the phase boundary between the paramagnetic and FM phases, whereas the white solid line ($J_{\pm,c} = 0.052$) indicates the transition from the FM to the QSI_0 phases in panel (b). Let us first define the total spinon QFI $f_Q(S_{\mathbf{q}}^{\pm}, T) = f_Q(S_{\mathbf{q}}^x + S_{\mathbf{q}}^y, T) = f_Q(S_{\mathbf{q}}^x, T) + f_Q(S_{\mathbf{q}}^y, T)$ under the XXZ model. Panels (a, b) and (c, d) show $f_Q(S_{\Gamma}^{\pm}, T)$ and $f_Q(S_{\Gamma'}^{\pm}, T)$ respectively at different values of $J_{\pm}$. Among

which, (a, c) are obtained from ED and (b, d) are from QMC on a $L = 4$ lattice ($4 \times L^3$ sites). We additionally computed the QFI obtained from GMFT and compared it with QMC simulations at $J_{\pm} = 0.045$ for system sizes $L = 3$ and $L = 4$. We find that, while QMC and ED results are consistent with each other, GMFT results are only off by a factor of approximately 6/7 compared to QMC and ED at low temperatures (see SM [101]). This is impressive given the mean-field nature of the GMFT. As such, we argue that the ED and GMFT results can be readily extended into the $J_{\pm} < 0$ regime.

We would like to first highlight that QFI shows non-trivial entanglement depth across various phases. For spin-1/2 systems, we find $n\text{QFI}(S_{\mathbf{q}}^{\pm}) = f_Q(S_{\mathbf{q}}^{\pm}, T)/2$ (see SM [101]). As shown in Fig. 1(a,b), $f_Q(S_{\Gamma}^{\pm}, T) \gtrsim 3$ in QSI_0 and $\gtrsim 5$ in the FM phase, implying at least 2- and 3-partite entanglement, respectively, in sharp contrast to the high-temperature paramagnet (PM), where $f_Q(S_{\Gamma}^{\pm}, T) = 0$. On the other hand, we see that $f_Q(S_{\Gamma}^{\pm}, T)$ of QSI_{π} is ~ 0 . We hereby stress that QFI is a mere lower bound of entanglement depth — such results do not reflect that QSI_{π} is trivially entangled nor that the FM phase is more entangled than that of the QSI_{π} phase. In fact, QFI is heavily dependent on the momentum positions. In principle, to aptly categorize the lower bound of the entanglement depth of a certain phase, one should rigorously search through the momentum space to find the largest QFI. One guiding principle for determining said momentum position is to examine the equal-time spin structure factor (ETSF), since the QFI density at $T \rightarrow 0$ becomes $\langle S_{-\mathbf{q}}^+ S_{\mathbf{q}}^- + S_{-\mathbf{q}}^- S_{\mathbf{q}}^+ \rangle / 2$. We demonstrate this point by presenting $f_Q(S_{\Gamma'}^{\pm}, T)$ in Fig. 1(c, d), where $\Gamma' = (4\pi, 4\pi, 0)$, at different values of $J_{\pm}$. This momentum position corresponds to a local maximum of the ETSF for QSI_{π} [94, 95] and, consequently, a local maximum for QFI at low temperatures [20, 45]. Indeed, here $f_Q(S_{\Gamma'}^{\pm}, T)$ of QSI_{π} is $\gtrsim 3$, whereas for QSI_0 and FM they are $\gtrsim 2$ and $\gtrsim 1$ respectively. From this, we can conclude that QSI_{π} is at least 2-partite entangled.

Beyond providing a lower bound on entanglement depth, the QFI is a channel-resolved measure of quantum coherence, and therefore depends explicitly on the choice of probe operator. For thermal states, the temperature kernel in Eq. S10 effectively suppresses quasi-static contributions with $\omega \ll T$, which are strongly thermally populated and typically correspond to slow, relaxational fluctuations that dominate classical noise in equal-time correlators and static susceptibilities [23, 31, 126]. In contrast, the QFI preferentially weights coherent dynamical fluctuations at $\omega \gtrsim T$ in the specific channel accessed by $S_{\mathbf{q}}^{\alpha}$, so changing α and $\mathbf{q}$ directly tunes which excitations contribute most strongly [23, 31, 127]. This built-in quantum selectivity and operator tunability make f_Q a sharp diagnostic of the thermal crossovers and phase boundaries. In particular, at finite temperature, $f_Q(S_{\Gamma}^{\pm}, T)$ faithfully tracks the two crossover temperatures of QSI_0 regime as shown in Fig. 2(a) for $J_{\pm} = 0.045$ and 0.05. From previous QMC calculations of the specific heat [52], it is known that as one cools down from the high-temperature paramagnetic (PM) phase,

the system first crosses over into the classical spin ice (CSI) at $T \sim 1$ before finally entering the QSI_0 regime at $T \sim |J_{\pm}^3|$. These crossover temperatures are marked by the two peaks in the specific heat. But these features are equally well represented in the QFI density. In Fig. 2(a), $f_Q(S_{\Gamma}^{\pm}, T)$ first starts increasing from 0 in the PM phase at $T \sim 1$ when we cross over into CSI. After which, $f_Q(S_{\Gamma}^{\pm}, T)$ plateaus until the second temperature scales $T \sim |J_{\pm}^3|$ where $f_Q(S_{\Gamma}^{\pm}, T)$ rises again as we cross over into the QSI_0 phase. Finally, $f_Q(S_{\Gamma}^{\pm}, T)$ reaches another plateau, which is consistent with the GMFT calculation (the green triangle in Fig. 2(a)) up to a factor of approximately $\sim 6/7$. Similarly, the phase boundary between the FM phase and the high-temperature PM is clearly delineated by the increase in $f_Q(S_{\Gamma}^{\pm}, T)$, as shown in Fig. 2(a). As such, the QFI also appears to be sensitive to this thermal phase transition [53].

On the other hand, the two temperature crossover scales are also clearly demarcated by QFI for QSI_{π} phase, as shown in Fig. 1(a,c): both $f_Q(S_{\Gamma}^{\pm}, T)$ and $f_Q(S_{\Gamma'}^{\pm}, T)$ increase near $T \sim 1$; at $T \sim |J_{\pm}|^3$, $f_Q(S_{\Gamma}^{\pm}, T)$ decreases while $f_Q(S_{\Gamma'}^{\pm}, T)$ increases for $J_{\pm} \lesssim -0.3$. At $J_{\pm} \sim -0.3$, the two temperature scales coalesce, as shown by the essentially flat QFI below the first temperature scale 1K in Fig. 2(a,b). The behavior is consistent with $\text{Ce}_2\text{Zr}_2\text{O}_7$ (best-fit $J_{\pm} \approx -0.3$ [78, 81, 87]), where only one peak is observed in specific heat measurement [87]. In contrasts, specific heat measurements on $\text{Ce}_2\text{Hf}_2\text{O}_7$ ($J_{\pm} \approx -0.125$ in the QSI scenario) [83–85, 128] show two well-separated peaks [84] mirrored by the predicted two inflection points of QFI, most pronounced in $f_Q(S_{\Gamma}^{\pm}, T)$ as shown in Fig. 2(a).

The evolution of QFI also strongly depends on the momentum position. The two aforementioned temperature scales manifest themselves in a strikingly different way when looking at $f_Q(S_{\Gamma'}^{\pm}, T)$ in Fig. 2(b) as opposed to $f_Q(S_{\Gamma}^{\pm}, T)$ in Fig. 2(a). As we cool down to the CSI regime from the high-temperature PM at the temperature scale J_z , $f_Q(S_{\Gamma'}^{\pm}, T)$ rapidly rises and eventually plateaus. But after the second temperature scale is met, $f_Q(S_{\Gamma'}^{\pm}, T)$ now dips instead of increasing, giving rise to this characteristic hump as shown in Fig. 2(b), which precisely marks the region of large critical fluctuations. This speaks to the versatility of QFI as a potentially powerful experimental probe — by adjusting the momentum positions, we can design probes that are sensitive to specific features of interest.

DISCUSSION

So far, we have considered the QFI of an abstract XXZ model. The same framework connects directly to real neutron scattering experiments once the microscopic nature of the local moments and their coupling to the neutron's magnetic moment are specified (see detailed derivation in SM. [101]). In particular, we present predictions in Fig. 3 for the Ce-based dipolar–octupolar pyrochlore $\text{Ce}_2\text{Zr}_2\text{O}_7$ [78, 87, 129]—a leading QSI candidate. While Ce ions carry both dipolar and oc-

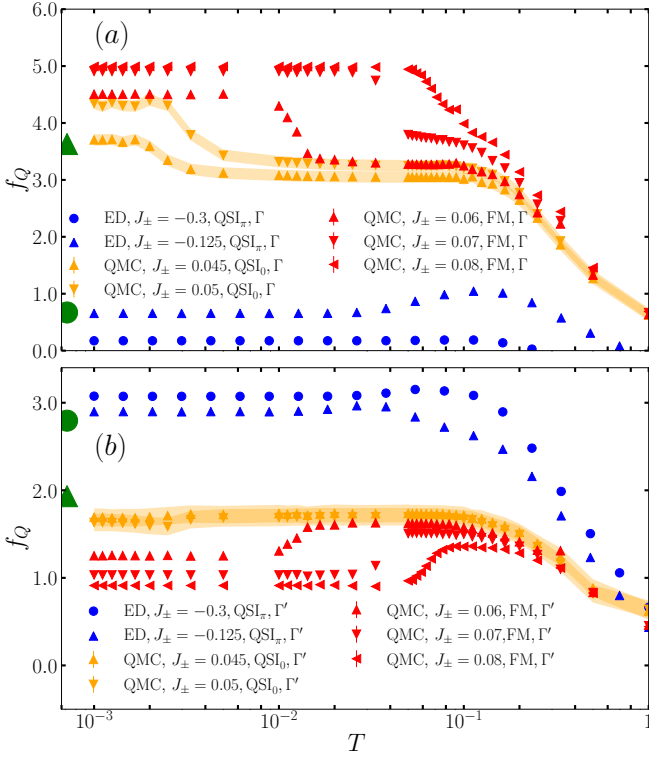


FIG. 2. **Temperature evolution of the QFI in different phases obtained with various computational methods.** The QFI $f_Q(S_q^\pm, T)$ are shown for (a) the $S^\pm$ channel at the Γ point, (b) the $S^\pm$ channel at the Γ' point. The blue disk points represent results from ED calculations with $J_\pm = -0.3$ and $J_\pm = -0.125$ in the QSI_π regime, while the orange triangle correspond to QMC simulations with $J_\pm = 0.045$ and $J_\pm = 0.05$ in the QSI_0 regime. The orange shaded areas accompanying the data highlight the crossover from paramagnetic regime to classical spin ice regime at $T \sim 1$ and that from the classical spin ice regime to QSI_0 at $T \sim |J_\pm^3|$. The red triangles indicate QMC results with $J_\pm$ ranging from 0.06 to 0.08 in the FM regime. The green circles near the y-axis indicate the GMFT calculation results at $J_\pm = -0.3$ (QSI_π) at zero temperature, while the green triangle corresponds to the result at $J_\pm = 0.045$ (QSI_0). The QMC, ED, and GMFT results are consistent (see SM [101] for details).

tupolar moments, neutron scattering primarily couples to the dipolar moments, which correspond to transverse spin components $S^\pm$ in the current model [75, 130]. Fig. 3 shows the corresponding QFI $f_Q(S_q^{\text{DO}}, T)$ that takes into account the transverse momentum projector in the inelastic neutron scattering.

In Fig. 3, panels (a,b) are obtained from QMC simulation for $L = 4$ and panels (c,d) are the ED results. One sees that in the global frame, the QFI again vividly captures the thermodynamic phase structure. The crossover temperature scale from paramagnetic region at high temperature ($T > J_z$) to the CSI and its plateau at intermediate temperature ($|J_\pm^3| < T < J_z$) and eventually the QSI_0 at low temperature ($T < |J_\pm^3|$) manifest in the both the ED and QMC data. The evolution of QFI closely resembles that of the thermal entropy measurements [52].

One important point is that, in principle, the bounds on $f_Q(S_q^{\text{DO}}, T)$ for an m -producible state are momentum depen-

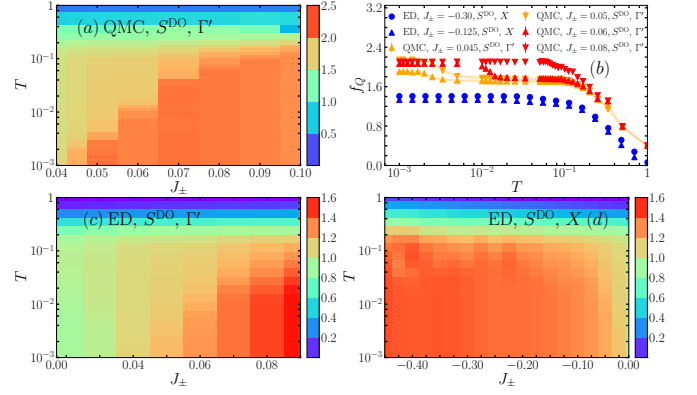


FIG. 3. **QFI in experimental coordinates for Cerium-based pyrochlore compounds.** Panel (a) presents a heat map of the QMC computed QFI $f_Q(S_q^{\text{DO}}, T)$, with $J_\pm$ ranging from 0.04 to 0.10. Panel (b) is the line cut of panel (a) from $J_\pm = 0.045$ to 0.08. The crossover temperature scales from the paramagnetic phase to CSI at $T \sim 1$ and from CSI to QSI_0 at $T \sim |J_\pm^3|$ are clearly manifest in the shaded data, highlighted in orange. Panel (c) shows ED results of $f_Q(S_q^{\text{DO}}, T)$ for $J_\pm$ between 0.01 and 0.08. Panel (d) focuses $f_Q(S_q^{\text{DO}}, T)$ with $X = (0, 0, 2\pi)$ in the range from -0.425 to 0.00 .

dent because of the transverse projector. But we can establish a $\mathbf{q}$ -independent bound as derived in SM [101], giving us the following relationship: $n\text{QFI}(S_q^{\text{DO}}) = \frac{3}{2}f_Q(S_q^{\text{DO}})$. As a result, values extracted directly from our computed data in a neutron scattering setting already certify nontrivial entanglement depth: $n\text{QFI}(S_q^{\text{DO}}, T) \gtrsim 2.2$ for 0-flux QSI and $n\text{QFI}(S_q^{\text{DO}}, T) \gtrsim 3.7$ for all-in-all-out order (FM in the local spin frame) respectively. At the $X = (0, 0, 2\pi)$ point, $n\text{QFI}(S_q^{\text{DO}}, T) \gtrsim 2.3$ certifies that the π -flux QSI is at least 3-partite entangled. Moreover, in principle, polarized neutron measurements with the non-spin-flip (NSF) and spin-flip (SF) channels are the optimal probes to find the best entanglement depth bound. As shown in SM [101], $n\text{QFI}(S_q^{(\text{NSF})}) = 3f_Q(S_q^{(\text{NSF})})$, therefore, even a modest $f_Q(S_q^{(\text{NSF})})$ can signal deep multipartite entanglement.

Based on the results and analysis above, we believe the QFI can actually offer richer information than initially anticipated. Besides providing insightful information about the entanglement depth, it can be used as both a thermal and dynamical probe in the investigation of quantum magnets, with quantum spin ice as an archetypal manifestation. Based on spin spectra in both local and global frames, the temperature and momentum dependence of QFI can faithfully trace out the ground state and finite-temperature phase boundaries in the QSI system, consistent with previous theoretical and numerical knowledge. These properties of QFI can bridge the fundamental entanglement concept to practical model computations and applications to experiments. In this way, other exotic quantum critical points, QSL phases, and their associated theory, computation, and experimental verification, could be expected to unite from the entanglement perspective.

Methods

The numerical results for the pyrochlore lattice XXZ model

are obtained from three methods: quantum Monte Carlo (QMC) simulations with a multi-directed loop (MDL) update scheme, exact diagonalization (ED), and gauge mean-field theory (GMFT) calculations. In addition, we transfer our observations to the neutron scattering section that connects directly to the real experiments. Here, we briefly introduce all three numerical methods and this transformation. The details of these methods are provided in the Supplemental Material [101].

Quantum Monte Carlo simulations with the multi-directed loop update The QMC simulations are performed based on the stochastic series expansion (SSE) framework [97, 98, 100], where the configuration space is constructed by expanding the partition function, $Z = \text{Tr}(e^{-\beta H})$, as a Taylor series into an operator string. We apply the diagonal update, directed loop (DL) update, and multi-directed loop (MDL) update schemes in our simulation. Our simulations are performed on system sizes $L = 3$ and 4. The MDL update is specifically designed to efficiently sample the highly frustrated (3+1)d configurational space of the pyrochlore lattice XXZ model, where the simplest quantum fluctuation that connects different classical spin-ice states involves six spin flips around a hexagon. This nature limits the sampling efficiency of the traditional DL update. To overcome this limitation, the MDL update allows the insertion of multiple operator pairs, thereby achieving efficient sampling and making the QSI₀ phase accessible (see SM [101] for details). In this update method, we first randomly select a hexagon in the configurational space and insert three pairs of operators, S_i^+ and S_i^- , on three alternate sites of the hexagon. Then, for each pair of operators, we randomly choose one operator to be the head of the loop, while the other operator becomes the tail. These three heads then propagate through the operator string until all of these heads meet their tails again at the same propagation step, at which point all loops are closed and the update process is completed.

In addition, we measure the imaginary time correlation function by tracing the evolution of the configuration space in the QMC simulation [122]. We then utilize the stochastic analytic continuation (SAC) scheme [116–120] to convert the imaginary time correlation function to a real-frequency dynamic structure factor.

Exact diagonalization The exact diagonalization scheme follows that of the microcanonical thermal pure quantum method (mTPQ) [102, 103]. The spectral function is obtained from an mTPQ representative state $|\psi_k\rangle$ using the broadened resolvent of a chosen operator O [131, 132],

$$G_O(z; \beta_k) = \frac{\langle \psi_k | O^\dagger (z - \tilde{H})^{-1} O | \psi_k \rangle}{\langle \psi_k | \psi_k \rangle}, \quad (5)$$

with $\tilde{H} \equiv H - E$ and $z = \omega + i\eta$. Here $\eta > 0$ controls the Lorentzian broadening, and the shift E fixes the absolute frequency reference. We choose E as the ground-state energy, obtained independently from a separate Lanczos run, so that the spectrum is reported with the correct energy offset. The

associated spectral density follows as

$$I_O(\omega; \beta_k) = -\frac{1}{\pi} \text{Im} G_O(\omega + i\eta; \beta_k). \quad (6)$$

Specifically, we perform the ED calculations on a 16-site periodic lattice, where four tetrahedra construct a large tetrahedron. Such a geometry respects the full symmetry of the pyrochlore lattice.

Gauge mean-field theory calculations The gauge mean-field theory (GMFT) calculations are performed to calculate the dynamical structure factor and QFI by canonically mapping the spin Hamiltonian to that of a lattice $U(1)$ gauge theory coupled to bosonic matter fields. We note that the results of GMFT align well with the QMC results with a factor of approximately 6/7 (see SM [101] for details).

Neutron scattering cross section The neutron scattering cross-section is proportional to

$$A^{\text{DO}}(\mathbf{q}, \omega) = \sum_{\mu, \nu} \left(\hat{\mathbf{z}}_\mu \cdot \hat{\mathbf{z}}_\nu - \frac{(\hat{\mathbf{z}}_\mu \cdot \mathbf{q})(\hat{\mathbf{z}}_\nu \cdot \mathbf{q})}{|\mathbf{q}|^2} \right) A_{\mu, \nu}^\pm(\mathbf{q}, \omega), \quad (7)$$

where μ and ν are the four sublattice indices of the pyrochlore lattice, and $A_{\mu, \nu}^\pm(\mathbf{Q}, \omega)$ is the spectral function obtained in the local frame. $\hat{\mathbf{z}}_\mu$ is the local z -axis direction of sublattice μ , which are given by

$$\begin{aligned} \mathbf{z}_0 &= \frac{1}{\sqrt{3}}(1, 1, 1) & \mathbf{z}_1 &= \frac{1}{\sqrt{3}}(1, -1, -1) \\ \mathbf{z}_2 &= \frac{1}{\sqrt{3}}(-1, 1, -1) & \mathbf{z}_3 &= \frac{1}{\sqrt{3}}(-1, -1, 1) \end{aligned} \quad (8)$$

Quantum Fisher information A meaningful QFI bound is defined with respect to a particular collective observable $O = \sum_i O_i$, such that $f_Q(O) \leq m(\Delta\lambda)^2$ when evaluated over an m -producible state and $\Delta\lambda$ is the spectral width of the individual observables O_i . However, in the main text, we use $S_{\mathbf{q}}^\alpha = \sum_i S_i^\alpha e^{i\mathbf{q} \cdot \mathbf{R}_i}$, which is not always Hermitian. Nevertheless, with the definition in Eq. (S10), we now show how to extract the essential properties of the QFI f_Q , such as the entanglement lower bound. A detailed derivation is provided in the Supplemental Material [101].

For a given operator $O_{\mathbf{q}} = \sum_i O_i e^{i\mathbf{q} \cdot \mathbf{R}_i}$, we use the Hermitian $O_{c, \mathbf{q}} = (O_{\mathbf{q}} + O_{\mathbf{q}}^\dagger)/2$ and the antihermitian $O_{s, \mathbf{q}} = (O_{\mathbf{q}} - O_{\mathbf{q}}^\dagger)/2i$ parts to construct a QFI matrix (QFIM) F [133–135], whose elements are specified by:

$$F_{ab}(T) := 4 \int d\omega \tanh\left(\frac{\omega}{2T}\right) (1 - e^{-\omega/T}) \mathcal{A}_{ab}(\mathbf{q}, \omega), \quad (9)$$

where

$$\mathcal{A}_{ab}(\mathbf{q}, \omega) = \frac{1}{2\pi N} \int dt e^{i\omega t} \langle O_{a, \mathbf{q}}(t) O_{b, \mathbf{q}}(0) \rangle, \quad (10)$$

with $a, b \in \{c, s\}$. $f_Q(O_{\mathbf{q}})$ defined in Eq. (S10) is equivalent to $\text{Tr}(F)$ (see SM [101] for more details). Therefore, since both diagonal elements are bounded by Eq. (2), for an

m -producible state, $f_Q(O_q) = f_Q(O_{c,q}) + f_Q(O_{s,q})$ is simply bounded by the sum of the individual QFI bounds. After detailed derivation, we find that:

$$f_Q(O_q, T) \leq m (\Delta(O_i))^2, \quad (11)$$

where $\Delta(O_i)$ is the spectral width of the operator O_i . When $O_q = S_q^\alpha$ ($\alpha \in \{x, y, z\}$) and $S_q^\pm$, the corresponding entanglement depth lower bound condition is that when

$$f_Q(S_q^\alpha, T) > m, \quad f_Q(S_q^\pm, T) > 2m, \quad (12)$$

the state is at least $(m + 1)$ -partite entangled.

Furthermore, we show how to extract essential properties of the QFI for the neutron scattering cross section via Eq. (7). This is noteworthy because there does not exist any collective operator, S^{DO} , such that its two-point correlation function, $\langle S^{\text{DO}\dagger} S^{\text{DO}} \rangle$, is equivalent to A^{DO} . Nevertheless, we can rewrite it as a sum of the QFI induced by the non-spin-flip (NSF) channel with underlying operator $S_q^{\text{NSF}} = \sum_{\mathbf{R}_\mu} (\hat{\mathbf{p}} \cdot \hat{\mathbf{z}}_\mu) \tau_{\mathbf{R}_\mu}^z e^{i\mathbf{q} \cdot \mathbf{R}_\mu}$ and the spin-flip (SF) channel with operator $S_q^{\text{SF}} = \sum_{\mathbf{R}_\mu} (\hat{\mathbf{v}} \cdot \hat{\mathbf{z}}_\mu) \tau_{\mathbf{R}_\mu}^z e^{i\mathbf{q} \cdot \mathbf{R}_\mu}$, where $\mathbf{p} \perp \mathbf{q}$ defines a neutron polarization vector and $\mathbf{v} = \mathbf{p} \times \mathbf{q}$ defines the other transverse direction. Therefore, $f_Q(S_q^{\text{DO}}) = f_Q(S_q^{\text{NSF}}) + f_Q(S_q^{\text{SF}})$, since $A^{\text{DO}}(\mathbf{q}) = \langle S_q^{\text{NSF}\dagger} S_q^{\text{NSF}} \rangle + \langle S_q^{\text{SF}\dagger} S_q^{\text{SF}} \rangle$ by construction. This defines $f_Q(S^{\text{DO}})$ as another sum witness whose individual components are well-bounded via Eq. (11). Thus, $f_Q(S_q^{\text{DO}})$ is bounded by the sum of the underlying NSF and SF channels. Applying Eq. (11) to $S_q^{(\text{NSF})}$, we arrive at

$$f_Q(S_q^{(\text{NSF})}, T) > m/3, \quad f_Q(S_q^{\text{DO}}, T) > 2m/3, \quad (13)$$

which implies that the state is at least $(m+1)$ -partite entangled. See SM [101] for a detailed derivation.

Data availability The data that support the findings of this study are provided at <https://doi.org/10.25442/hku.32404548>.

Code availability The main codes in this paper are available at <https://doi.org/10.25442/hku.32412273>.

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Authors contributions Z.Y.M. and Y.B.K. supervised the project. C.K.Z. developed the multi-directed loop update scheme and performed the QMC simulations. Z.Z. and F.D. performed the ED calculations and the GMFT calculations. All authors contributed to writing the manuscript.

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Supplemental Material for “Quantum Fisher Information as a Thermal Probe in Frustrated Magnets through Insights from Quantum Spin Ice”

The Supplemental Material provides details on quantum Monte Carlo simulations using a multi-directed loop (MDL) update scheme, specifically designed for quantum spin ice systems, as well as on the measurement of quantum Fisher information and other physical observables therein. It also includes detailed benchmarking between QMC, exact diagonalization, and gauge mean field theory calculations. We also discuss the conversion of local frame QFI data to that obtained from the neutron scattering cross section of dipolar-octupolar QSI material candidates in the global frame.

I. MULTI-DIRECTED LOOP QMC SCHEME FOR QUANTUM SPIN ICE MODEL

In this section, we briefly introduce the stochastic series expansion quantum Monte Carlo (SSE-QMC) simulation on the pyrochlore lattice [97, 98, 100]. The basic idea is to construct the configuration space by expanding the partition function, $Z = \text{Tr}(e^{-\beta H})$, as a Taylor series into an operator string. We then sample this configuration space by proposing updates to the operator string, which fulfill the balance condition[137], $\sum_i W(A_i)P(A_i \rightarrow B) = W(B)$, or even the detailed balance condition, $W(A)P(A \rightarrow B) = W(B)P(B \rightarrow A)$. Here, A_i are all possible configurations that can be updated to configuration B . $W(A)$ is the weight of configuration A , and $P(A \rightarrow B)$ is the transition probability from configuration A to B . To do so, one of the widely used methods is the directed loop (DL) update algorithm.

Within the SSE-QMC framework, the DL update algorithm is a highly efficient method for simulating quantum spin systems by inserting a pair of operators (e.g., S_i^+ and S_i^-), the configuration temporarily moves into an extended configuration space with these two operators evolving, and returns to the Hilbert space when the two operators meet and annihilate. However, on the pyrochlore lattice within the spin-ice regime ($0 < J_{\pm} < 0.052$), the DL update fails to drive the system into the quantum spin ice (QSI) regime. These are plotted as orange points in Fig. S2(a-c). This failure can be understood from the perspective that first-order perturbations vanish, while third-order processes persist in the spin-ice regime [46]. Consequently, the simplest quantum fluctuation that connects different classical spin-ice states involves six spin flips around a hexagon, which is challenging to realize by inserting only a single operator pair. To address this problem, we develop a multi-directed loop (MDL) update that allows the insertion of multiple operator pairs, enabling efficient sampling in the QSI regime.

In our simulation, we first decompose the Hamiltonian into a sum of tetrahedron operators (see Fig.S1(a)), which is given

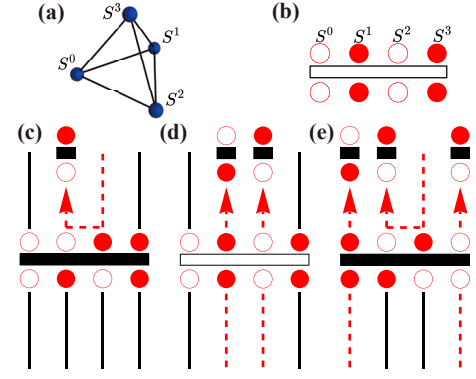


FIG. S1. Schematic diagram of (a) a tetrahedron in the pyrochlore lattice and (b) the corresponding four leg operator vertex in the sampling space. (c) Illustration of the single loop update, (d) bi-loop update, and (e) tri-loop update in the multi-directed loop algorithm.

by

$$\begin{aligned}
 H &= \sum_t H_{t,0} + \sum_t H_{t,1}, \\
 H_{t,0} &= J_z \sum_{\langle i,j \rangle \in \boxtimes} S_i^z S_j^z + H_{c,0}, \\
 H_{t,1} &= -J_{\pm} \sum_{\langle i,j \rangle \in \boxtimes} (S_i^+ S_j^- + S_i^- S_j^+).
 \end{aligned} \tag{S1}$$

Here, t denotes a tetrahedron ($\boxtimes$) in the pyrochlore lattice and $\langle i, j \rangle \in \boxtimes$ represents the nearest-neighbor pairs within the tetrahedron t . To ensure that all matrix elements are non-negative, we introduce a constant shift $H_{c,0}$ to the diagonal operator, which is $H_{c,0} = 3J_z/2$. Therefore, for a given tetrahedron state $|\alpha_i\rangle = |S_1^z, S_2^z, S_3^z, S_4^z\rangle$, the matrix elements of the operators are $w_{i,j,x} = \langle \alpha_j | H_{t,x} | \alpha_i \rangle$ with $x = 0, 1$. And, the weight of a given operator string is

$$W = \frac{\beta^n (L-n)!}{L!} \prod_{p=1}^n w_{i_p, j_p, x_p}, \tag{S2}$$

where L refers to the cut-off of the expansion and n is the number of operators in the string. In practice, each $H_{t,x}$ is represented by a 8-leg vertex, as illustrated in Fig. S1(b). We employ the diagonal update, DL update, and MDL update to sample the configuration space efficiently. Also, we apply a thermal annealing process from inverse temperature $\beta = 1$ to $\beta = 1000$ with step $\delta\beta = 1$. At each $\beta = \frac{1}{T}$, we perform 10,000 steps of warming and 10,000 steps of measurement. The details of these updates are as follows.

1. Diagonal update

In the diagonal update, we follow the detailed balance condition and apply the Metropolis algorithm. We sweep through the operator string and attempt to insert or remove diagonal

operators $H_{t,0}$ at each time slice. In the insertion process, we randomly select a tetrahedron t and propose to insert the diagonal operator $H_{t,0}$. For the removal process, each diagonal operator are proposed to be removed. Therefore, the acceptance probabilities for insertion is given by

$$P_{\text{accept}}^{\text{insert}} = \min \left[1, \frac{N_t \beta w_{i,i,0}}{L - n} \right], \quad (\text{S3})$$

and for removal is

$$P_{\text{accept}}^{\text{remove}} = \min \left[1, \frac{L - n + 1}{N_t \beta w_{i,i,0}} \right], \quad (\text{S4})$$

with $w_{i,i,0}$ denotes the weight of $H_{t,0}$ at this time slice, and N_t represents the total number of tetrahedron (∇).

2. Directed loop update

In the DL update, we follow the balance condition and utilize the update scheme proposed in Ref. [137] to achieve the highest efficiency sampling as we know [123, 138]. In practice, we first randomly select a vertex and one of its eight legs to insert a pair of operators, S_i^+ and S_i^- . The selection probability is $P_{\text{select}} = 1/(N_v \times 8)$, where N_v is the total number of vertices in the given operator string. Then, we choose one of these pairs of operators to be the head, while the other operator becomes the tail. The head then propagates through the operator string by entering and exiting vertices until it meets the tail again, at which point the loop is closed. The propagation process is illustrated in Fig.S1(c). When the head enters a vertex through one leg (l_a), the exiting leg (l_b) is chosen based on a set of exit probabilities P_{exit} , which update the operator from w_a to w_b . The exiting leg is selected according to the probability so that the balance condition (or the detailed balance condition) is satisfied.

In our simulation, the exit probabilities P_{exit} are determined by following the balance condition and utilizing the algorithm proposed in Ref. [137] to minimize the average rejection rate, which means the head exits from the entrance leg without changing the vertex. The exit probabilities are given by

$$\begin{aligned} P_{\text{exit}}(l_a \rightarrow l_b) &= \frac{1}{w_a} \max[0, \min[\Delta_{ab}, w_a + w_b - \Delta_{ab}, w_a, w_b]], \\ \Delta_{ab} &= S_a - S_{a-1} + w_1, 1 \leq a, b \leq n, \\ S_i &= \sum_{k=1}^i w_k, 1 \leq i \leq n, \\ S_0 &= S_n. \end{aligned} \quad (\text{S5})$$

Here, $n = 8$ is the total number of candidate exit legs. In practice, we enumerate all nonzero vertices and iterate over all possible entrance legs to calculate and store the corresponding exit probabilities. For each vertex and entrance leg, we identify all candidate exiting legs (n in total) and their associated weights w_i , which are sorted in descending order. The

sums $\{S_i\}$ are then computed for these candidates and used to determine the exit probabilities $P_{\text{exit}}(l_a \rightarrow l_i)$. All results are precomputed and stored before the simulation. During the simulation, the exiting leg l_b is selected according to these stored probabilities. Such an update method achieves highly efficient sampling in various quantum spin systems, including the Balents-Fisher-Girvin model [123, 138]. However, in the spin-ice regime, the DL update still fails to drive the system into the QSI phase, as shown by the orange points in Fig. S2(a-c).

3. Multi-Directed loop update

Inspired by the nature that the simplest quantum fluctuation that connects different classical spin-ice states involves six spin flips around a hexagon, we develop a MDL update that allows the insertion of multiple operator pairs, enabling efficient access to the QSI regime. In this update method, we first select a hexagon in the operator string and insert three pairs of operators, S_i^+ and S_i^- , on three alternate sites of the hexagon. Then, for each pair of operators, we randomly choose one operator to be the head, while the other operator becomes the tail. The three heads then propagate through the operator string by entering and exiting vertices until all of these heads meet the tail again at the same propagation step, at which point all the loops are closed. Different from the single loop update, the propagation process of the three heads is more complicated, including one head entrance and exit a vertex (single loop update in Fig.S1(c)), two heads enter and exit a vertex (bi-loop update in Fig.S1(d)), and three heads enter and exit a vertex (tri-loop update in Fig.S1(e)).

To show that MDL update fulfills the detailed balance condition, we follow the worm-antiworm construction principle[97]. In MDL update, the corresponding transition probability from configuration A to B is given by

$$P(A \rightarrow B) = P_{\text{insert}} \prod_{i=1}^{N_m} \prod_{j=1}^{n_v} P_{\text{prop}}(v_{i,j} \rightarrow v_{i+1,j}), \quad (\text{S6})$$

where P_{insert} is the probability of inserting multiple operator pairs, N_m is the number of propagation steps for these operator pairs when all loops are closed. And n_v is the number of vertices visited by the heads (for example, two heads entering one vertex as in Fig. S1(d), or three heads as in Fig. S1(e)). $P_{\text{prop}}(v_{i,j} \rightarrow v_{i+1,j})$ is the propagation probability for updating the j -th visited vertex from $v_{i,j}$ to $v_{i+1,j}$, corresponding to the choice of exit legs for all heads visiting this vertex. This propagation probability is determined by listing all candidate vertices that can be updated from $v_{i,j}$, and solving for the probabilities that satisfy the detailed balance condition, $P_{\text{prop}}(v_{i,j} \rightarrow v_{i+1,j})W(v_{i,j}) = P_{\text{prop}}(v_{i+1,j} \rightarrow v_{i,j})W(v_{i+1,j})$. And the transition probability for the antiworm process is given by

$$P(B \rightarrow A) = P'_{\text{insert}} \prod_{i=1}^{N_m} \prod_{j=1}^{n_v} P_{\text{prop}}(v_{N_m-i+1,j} \rightarrow v_{N_m-i,j}), \quad (\text{S7})$$

therefore, we have

$$\begin{aligned} \frac{P(A \rightarrow B)}{P(B \rightarrow A)} &= \frac{P_{\text{insert}}}{P'_{\text{insert}}} \prod_{i=1}^{N_m} \prod_{j=1}^{n_v} \frac{W(v_{i+1,j})}{W(v_{i,j})} \\ &= \frac{P_{\text{insert}}}{P'_{\text{insert}}} \frac{W(B)}{W(A)}. \end{aligned} \quad (\text{S8})$$

The MDL update is completed only when all three heads meet a different tail at one propagation step, ensuring that the inverse process is identical to the forward process and the insertion probabilities satisfy $P_{\text{insert}}/P'_{\text{insert}} = 1$. Therefore, the MDL update fulfills the detailed balance condition. Note that if one of the heads meets its tail and closes the corresponding loop while the other heads continue to propagate, the inverse process differs from the forward process, and the corresponding update violates the detailed balance condition. In practice, before the simulation, we enumerate all nonzero vertices and iterate over all possible entrance cases, including single-, bi-, and tri-loop cases (see Fig.S1(c-e)), to calculate and store the propagation probability for each exit case. For each vertex, entrance head number, and entrance leg, we identify the propagation probability for each candidate exit leg case via Eq. (S5). All probabilities are precomputed and stored prior to the simulation.

It is worth mentioning that such a MDL update enlarged the middle extended configuration space compared to the DL update, which is beneficial for sampling the QSI regime. However, it also increases the computational cost per update. Assuming that the heads appear in each vertex leg with equal probability after a long propagation, then the probability of the heads meeting the tails and closing the loops is $3!/(4N_v)^3$ with $N_v \sim L^3\beta$. Here, L refers to the system size. In construct, it is $1/(4N_v)$ for the DL update. Therefore, it is harder to close the loops for the MDL update, and the computational cost per update of MDL is roughly $(L^3\beta)^2$ times that of the DL update. This cost difference becomes even more significant in the FM phase. Therefore, we utilize both MDL and DL update to simulate the system in the QSI regime ($J_{\pm} = 0.04, 0.045, 0.05$), while only DL in the FM regime ($J_{\pm} = 0.06, 0.07, 0.08, 0.09, 0.10$).

4. Measurement

Here, we briefly introduce the measurement results of physical observables in our simulation. We first measure the energy per site e and compare the results obtained with and without MDL updates in the QSI regime, as shown in Fig. S2(a-c) for $J_{\pm}$ ranging from 0.04 to 0.05. The orange points correspond to the case without MDL update, while the blue points are with the MDL update. In all three cases, the MDL update successfully drives the system into the QSI regime at low temperature, as evidenced by a clear energy decrease, while the DL update does not. Specifically, in Fig. S2(b), a distinct energy drop from -0.2547 to -0.2550 at $T \sim 0.0011$ is observed in the MDL update results. This small energy

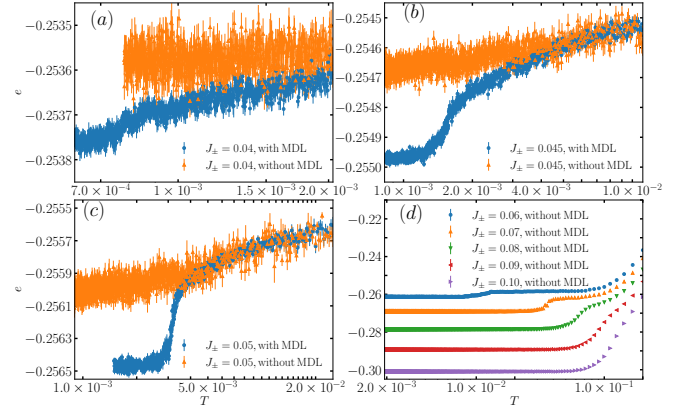


FIG. S2. The energy per site e as a function of temperature T at (a) $J_{\pm} = 0.04$, (b) $J_{\pm} = 0.045$, (c) $J_{\pm} = 0.05$, and (d) $J_{\pm}$ ranging from 0.06 to 0.10 with an increment of 0.01. In (a-c), the blue points are obtained from SSE-QMC simulation with MDL update, and the orange lines are without MDL update. In (d), all points are obtained from SSE-QMC simulation without MDL update.

drop corresponds to the crossover from classical spin ice (CSI) to QSI and is consistent with the theoretical prediction $T \sim 12J_{\pm}^3 \approx 0.0011$ and previous calculations [56]. In contrast, the DL update results remain around -0.2547 with only a slight decreasing tendency as temperature decreases, indicating that the DL update fails to sample the QSI regime efficiently and becomes trapped in a local minimum. Also, the energy density in the FM phase is shown in Fig. S2(d) for $J_{\pm}$ ranging from 0.06 to 0.10. In this regime, the DL update efficiently samples the configuration space and drives the system into the FM phase at low temperature.

Then, we measure the imaginary-time spin-spin correlation functions,

$$\begin{aligned} G^{\pm} &= \frac{1}{2N} \sum_{\gamma, \nu} \langle S_{-\mathbf{q}, \gamma}^+(\tau) S_{\mathbf{q}, \nu}^-(0) + S_{-\mathbf{q}, \gamma}^-(\tau) S_{\mathbf{q}, \nu}^+(0) \rangle, \\ G^{zz} &= \frac{1}{N} \sum_{\gamma, \nu} \langle S_{-\mathbf{q}, \gamma}^z(\tau) S_{\mathbf{q}, \nu}^z(0) \rangle, \end{aligned} \quad (\text{S9})$$

where τ is the imaginary time, γ and ν are sublattice indices, and N is the total number of sites. The $G^{\pm}$ correlation is measured by tracing the head's path in the DL update process [122]. The corresponding spectral function is then obtained via the stochastic analytic continuation (SAC) method [52, 120, 122, 123, 139]. Finally, the QFI density f_Q is calculated by integrating the spectral function according to Eq. (4) in the main text. Here, we focus on the QFI density at $\Gamma = (0, 0, 0)$, $\Gamma' = (4\pi, 4\pi, 0)$ and $X = (0, 0, 2\pi)$ in both the $S^{\pm}$ and S^z channel. The results for $f_Q(S_{\Gamma}^{\pm}, T)$ are shown in Fig. S3: panels (a) and (b) display $f_Q(S_{\Gamma}^{\pm}, T)$ for $J_{\pm}$ in the QSI regime (0.04 to 0.05) and FM regime (0.06 to 0.10), respectively. Panels (c) and (d) present heat maps of $f_Q(S_{\Gamma}^{\pm}, T)$ and energy density e as functions of temperature T and $J_{\pm}$ from 0.04 to 0.10. These two observations agree well, but the crossover signal in the QFI density is more significant.

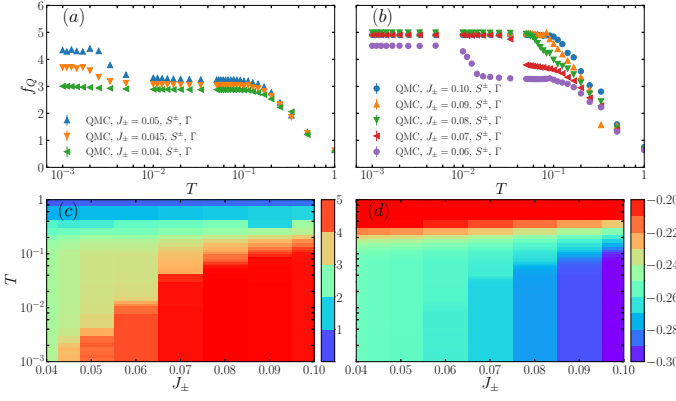


FIG. S3. The QFI density f_Q in $S^\pm$ channel at Γ as a function of temperature T with $J_\pm$ (a) ranging from 0.04 to 0.05 with an increment of 0.005; (b) ranging from 0.06 to 0.10 with an increment of 0.01. (c) The QFI density heat map as a function of temperature T and $J_\pm$, ranging from 0.04 to 0.10, while (d) is that of the energy density e heat map.

Fig. S4 presents the QFI density $f_Q(S_q^\pm, T)$ at the X and Γ' points. Panels (a) and (b) show $f_Q(S_X^\pm, T)$ and $f_Q(S_{\Gamma'}^\pm, T)$ in the QSI regime, while panels (c) and (d) display the corresponding results in the FM phase. The heat maps in panels (e) and (f) illustrate $f_Q(S_X^\pm, T)$ and $f_Q(S_{\Gamma'}^\pm, T)$ as functions of temperature T and $J_\pm$ from 0.04 to 0.10. In combination with Fig. S3, we find that $f_Q(S_X^\pm, T)$ exhibits the strongest signal in the $S^\pm$ channel, while $f_Q(S_{\Gamma'}^\pm, T)$ captures the critical fluctuations. Therefore, we focus on these two momentum points in the main text and include the results at X in the Supplemental Material for completeness.

Meanwhile, we also measure the QFI density $f_Q(S_q^z, T)$ in the S^z channel, as shown in Fig. S5. Panels (a) and (b) illustrate $f_Q(S_X^z, T)$ and $f_Q(S_{\Gamma'}^z, T)$ in the QSI regime, while panels (c) and (d) show the corresponding results in the FM phase. The heat maps of $f_Q(S_X^z, T)$ and $f_Q(S_{\Gamma'}^z, T)$ as functions of temperature T and $J_\pm$ from 0.04 to 0.10 are presented in panels (e) and (f), respectively. In panel (f), the $f_Q(S_{\Gamma'}^z, T)$ remains small for high temperature case, including the CSI regime. When the temperature is lowered upon entering the QSI regime, $f_Q(S_{\Gamma'}^z, T)$ increases and reaches a plateau. This behavior is consistent with the fact that the S^z channel captures the photon sector which emerges in the QSI regime. However, compared to the $S^\pm$ channel (the spinon sector), the QFI density in the S^z channel captures a similar picture but with a much smaller value, which is why we only include these results in the Supplemental Material for completeness.

Moreover, we compare the energy density e and QFI density f_Q between $L = 3$ (without MDL) and $L = 4$ (with MDL) QMC simulations, as shown in Fig. S6. The parameters are $J_\pm = 0.045$ and $J_z = 1$. The results for these two system sizes are consistent, indicating that finite-size effects are not significant in our simulation.

Finally, we compare the QFI density with the static spin

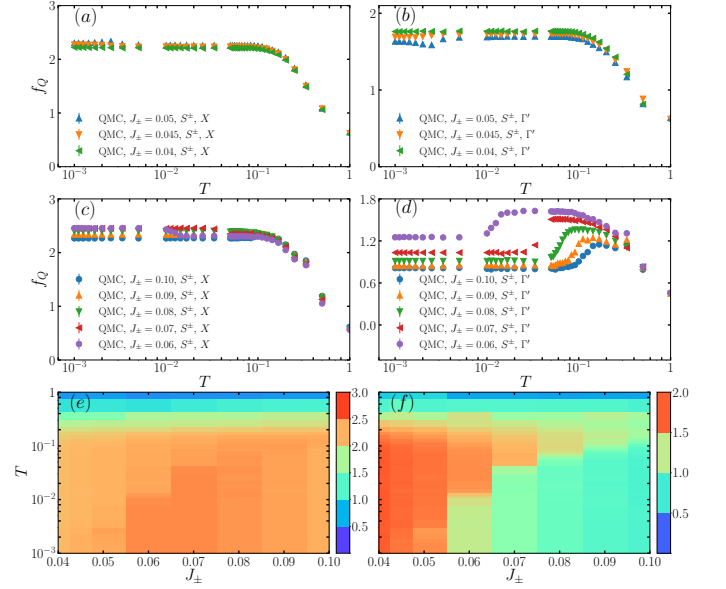


FIG. S4. The QFI density f_Q in $S^\pm$ channel (a, c, e) at X and (b, d, f) at Γ' as a function of temperature T and $J_\pm$. (a, b) illustrate f_Q with $J_\pm$ ranging from 0.04 to 0.05 with an increment of 0.005 while (c, d) is that of $J_\pm$ ranging from 0.06 to 0.10 with an increment of 0.01. (e, f) are the QFI density heat maps as a function of temperature T and $J_\pm$ ranging from 0.04 to 0.10.

structure factor (SSSF). By defining the kernel function $\ker(\omega, T)$, the QFI density (Eq. (4) in the main text) can be rewritten as:

$$f_Q(S_q^\alpha, T) = 4 \int_0^\infty d\omega A^\alpha(\mathbf{q}, \omega) \ker(\omega, T), \quad (S10)$$

$$\ker(\omega, T) = \tanh\left(\frac{\omega}{2T}\right) (1 - e^{-\omega/T}),$$

where the smooth thermal kernel $\ker(\omega, T)$ approaches unity only as $T \rightarrow 0$. This kernel function in QFI weights spectral contributions fundamentally differently from equal-time observables, such as the SSSF. The QFI kernel suppresses quasi-elastic thermal weight and highlights coherent quantum transitions, whereas the kernel for SSSF is simply unity and thus includes all spectral weight equally. This distinction is directly visible in our numerical results (see Fig. S7), which compare QFI and SSSF for two representative cases at $J_\pm = 0.045$: the $S^\pm$ channel at the Γ point and the S^z channel at the Γ' point. Panels (a-b) and (d-e) show the dynamical structure factor at two temperatures (blue curves from QMC), together with the corresponding QFI kernel (red curves). To illustrate the interplay between the dynamical structure factor and the kernel function, we have normalized the dynamical structure factor so that its maximum value equals 1, and the orange shaded regions indicate the spectral weight contributing to QFI. Panels (c) and (f) then show the temperature dependence of QFI (orange) and $4 \times \text{SSSF}$ (blue) for the $S^\pm$ channel at the Γ point and for the S^z channel at the Γ' point, respectively. These results reveal the distinct temperature dependence between QFI and the $4 \times \text{SSSF}$.

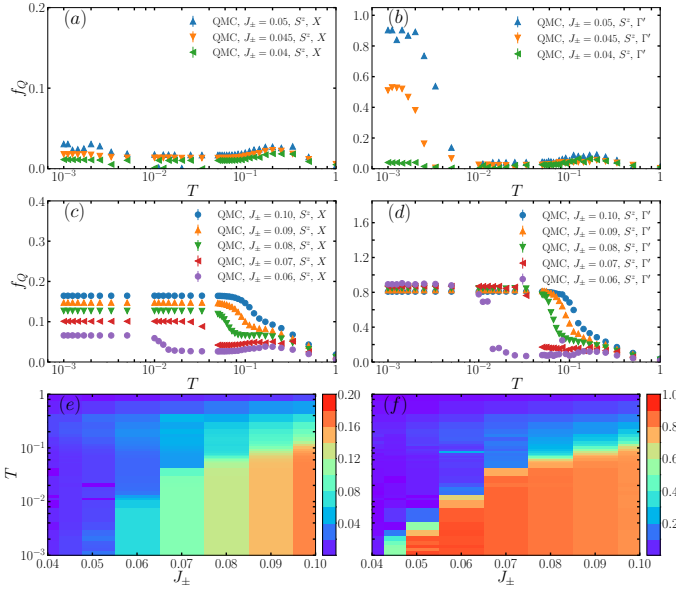


FIG. S5. The QFI density f_Q in S^z channel (a, c, e) at X and (b, d, f) at Γ' as a function of temperature T and $J_\pm$. (a, b) illustrate f_Q with $J_\pm$ ranging from 0.04 to 0.05 with an increment of 0.005 while (c, d) is that of $J_\pm$ ranging from 0.06 to 0.10 with an increment of 0.01. (e, f) are the QFI density heat maps as a function of temperature T and $J_\pm$ ranging from 0.04 to 0.10.

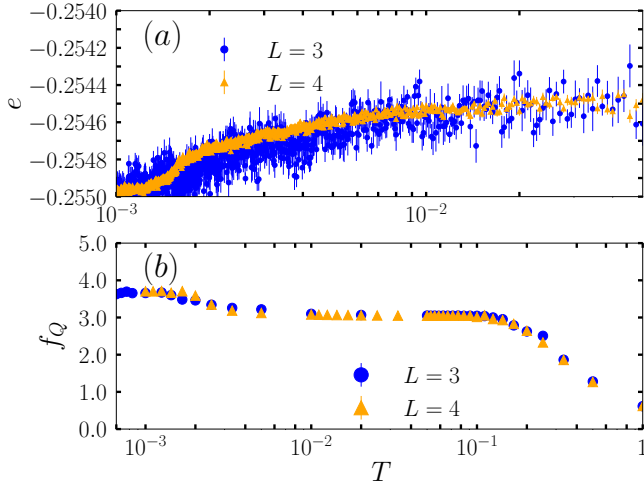


FIG. S6. The comparison of the energy density (e) and density of QFI (f_Q) between the $L = 3$ and $L = 4$ obtained from the QMC simulations. The parameters are $J_\pm = 0.045$ and $J_z = 1$.

In the spinon-dominated case $O(\mathbf{q}) = S_{\Gamma'}^\pm$, panel (c) demonstrates that QFI and four times the $4\times\text{SSSF}$ differ significantly in the high-temperature regime ($T > 10^{-1}$), but converge in the low-temperature regime ($T < 10^{-1}$). This behavior arises because the peak in the dynamical structure factor is located at $\omega \approx 1$ [Fig. S7(a-b)]. When T is sufficiently low (roughly $T \lesssim 10^{-1}$), the kernel effectively encompasses the entire peak, causing QFI and $4\times\text{SSSF}$ to become similar. At higher temperatures, however, the QFI kernel selectively reshapes which

parts of the spectrum contribute, resulting in distinct temperature dependence between QFI and $4\times\text{SSSF}$. This illustrates that, despite the expected low- T convergence, QFI and $4\times\text{SSSF}$ remain fundamentally different in their sensitivity to spectral features across temperatures.

A sharper and physically most important distinction emerges in the photon sector. Choosing $O(\mathbf{q}) = S_{\Gamma'}^z$ probes emergent gauge (photon) excitations. Here, the relevant spectral peak lies at very low frequency, $\omega \approx 0.005$ [Fig. S7(e)]. As a result, the temperature kernel ensures that QFI remains essentially zero over a broad temperature range and turns on only upon cooling into the photon energy window [Fig. S7(f)], thereby cleanly capturing the onset of the photon mode, a defining hallmark of QSI. By contrast, the SSSF remains substantial throughout because it is dominated by thermally activated quasi-elastic low-frequency weight, and thus does not provide a comparably clean discriminator for the photon sector. This operator-resolved selectivity is precisely the added value of QFI: it isolates the onset of coherent spinonic and photonic modes and yields a direct, physically transparent diagnostic of the QSI regime that static equal-time probes can obscure.

II. EXACT DIAGONALIZATION

The exact diagonalization scheme follows that of the microcanonical thermal pure quantum method (mTPQ) [102, 103]. We first construct a state at $T \rightarrow \infty$, $|\psi_0\rangle = \sum_{i=1} c_i |i\rangle$, such that $\{c_i\}$ is a set of random complex number and normalized $\sum_i |c_i|^2 = 1$. Then for a given Hamiltonian H , the k^{th} TPQ state is constructed iteratively via

$$|\psi_k\rangle = (L - H)|\psi_{k-1}\rangle, \quad (\text{S11})$$

where L is some constant value greater than the largest eigenvalue of the Hamiltonian. We can then estimate the energy and the inverse temperature via:

$$E_k = \frac{\langle \psi_k | H | \psi_k \rangle}{\langle \psi_k | \psi_k \rangle} \quad (\text{S12})$$

and

$$\beta_k = \frac{2k}{L - E_k}, \quad (\text{S13})$$

for the k^{th} iteration.

The microcanonical TPQ construction generates a sequence of pure states $\{|\psi_k\rangle\}$ whose energy distribution concentrates sharply around a typical energy E_k in the thermodynamic limit [102]. In the iterative update of Eq. (S11), successive applications of the filtering operator progressively enhance the weight of lower-energy components, thereby lowering E_k and scanning toward lower effective temperatures. As shown by Sugiura and Shimizu, expectation values of few-body observables evaluated in a single TPQ state self-average to the corresponding ensemble values, with relative

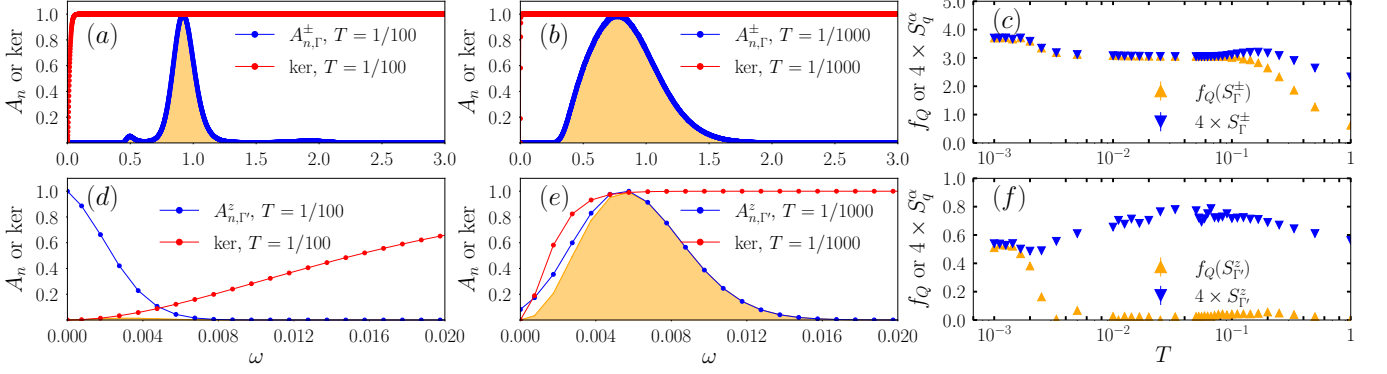


FIG. S7. The comparison between QFI and static structure factor. Panel (a-c) shows the dynamical and static structure factor and the QFI for $J_{\pm} = 0.045$ in $S^{\pm}$ channel at Γ point. Panel (d-f) shows the same quantities for $J_{\pm} = 0.045$ in the $S^{\pm}$ channel at the Γ' point. Panels (a) and (d) show the dynamical structure factor, normalized such that its highest value equals 1, at temperature $T = 1/100$, while panels (b) and (e) show that at $T = 1/1000$. In these four panels, the blue lines are the spectrums obtained from the QMC simulations, while the red lines are the corresponding kernel functions for the QFI calculation. And the orange shaded areas represent the contribution to the QFI. Panels (c) and (f) show the temperature dependence of the QFI (orange line), and the static structure factor (blue line) at the same momentum and channel.

fluctuations vanishing with increasing system size. Consequently, finite-temperature observables can be obtained efficiently from $\langle \psi_k | O | \psi_k \rangle / \langle \psi_k | \psi_k \rangle$ at the k corresponding to the desired temperature, without explicitly summing over the partition function.

We compute finite-temperature spectra from an mTPQ representative state $|\psi_k\rangle$ using the broadened resolvent of a chosen operator O [131, 132],

$$G_O(z; \beta_k) = \frac{\langle \psi_k | O^\dagger (z - \tilde{H})^{-1} O | \psi_k \rangle}{\langle \psi_k | \psi_k \rangle}, \quad (\text{S14})$$

with $\tilde{H} \equiv H - E$ and $z = \omega + i\eta$. Here $\eta > 0$ controls the Lorentzian broadening, and the shift E fixes the absolute frequency reference. We choose E as the ground-state energy, obtained independently from a separate Lanczos run, so that the spectrum is reported with the correct energy offset. The associated spectral density follows as

$$I_O(\omega; \beta_k) = -\frac{1}{\pi} \text{Im} G_O(\omega + i\eta; \beta_k). \quad (\text{S15})$$

To evaluate G_O , we introduce the normalized Krylov seed and its weight,

$$|f_0\rangle = \frac{O|\psi_k\rangle}{\|O|\psi_k\rangle\|}, \quad \mu_0 \equiv \frac{\|O|\psi_k\rangle\|^2}{\langle \psi_k | \psi_k \rangle}, \quad (\text{S16})$$

and perform Lanczos tridiagonalization of $\tilde{H}$ starting from $|f_0\rangle$. This produces the usual tridiagonal representation specified by diagonal elements $\{a_n\}$ and off-diagonal elements $\{b_{n+1}\}$ [132, 140, 141].

In the resulting Krylov basis, the matrix element of the resolvent admits the standard continued-fraction approxima-

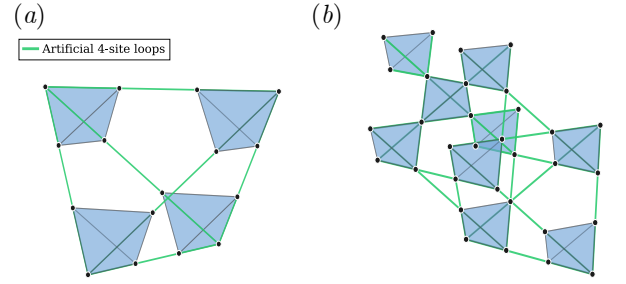


FIG. S8. (a) 16 sites cubic cluster setup for exact diagonalization (b) 32 sites $2 \times 2 \times 2$ cluster geometry. highlighted green edges are examples of the spurious 4-site loop enabled by finite size effects

tion [140, 141],

$$G_O(z; \beta_k) \approx \mu_0 \frac{1}{z - a_0 - \frac{b_1^2}{z - a_1 - \frac{b_2^2}{z - a_2 - \ddots}}}, \quad (\text{S17})$$

which we truncate after n_L Lanczos iterations. Combining Eqs. (S15) and (S17) then yields $I_O(\omega; \beta_k)$, giving us the desired spectral function. We choose a Krylov dimension of 800 for careful convergence of the spectral function.

It is important to distinguish two separate sources of uncertainty in our finite- T mTPQ calculations: (i) physical finite-size effects associated with working on a finite, symmetry-respecting pyrochlore cluster, and (ii) statistical (typicality) fluctuations associated with estimating thermal quantities using a finite number of random initial vectors. The mTPQ method provides an unbiased estimator of the finite- N canonical (or microcanonical) ensemble, but it does not remove physical finite-size effects, which can remain significant at low temperature when the low-energy spectrum is sparse.

For the pyrochlore lattice, $N = 16$ is the largest symmetry-respecting cluster that is feasible within our present finite-temperature dynamical workflow. The next symmetry-compatible cluster has $N = 32$ sites and is numerically intractable here. More importantly, the dominant finite-size artifact in the quantum spin-ice regime is not a smooth $1/N$ correction, but a geometry-enabled process tied to the short linear extent of these clusters: periodic boundary conditions permit an additional four-site exchange within the spin-ice manifold along the shortest edges. This modifies the effective low-energy dynamics by enhancing the characteristic photon scale to order $9J_{\pm}^2/J_{zz}$, instead of the thermodynamic-limit ring-exchange scale of order $12J_{\pm}^3/J_{zz}^2$ [46, 48, 69, 104, 142]. As shown in Fig. S8, this mechanism is a property of the cluster geometry (short edges spanning two tetrahedra) and therefore, is not expected to be removed simply by moving to the next symmetry-respecting cluster, where the same short-edge structure persists, and such processes remain allowed. The main consequence is a renormalization of the absolute photonic scale and hence a shift of the photon-related crossover temperature on finite clusters.

Despite this known artifact, the quantities we focus on are still well-motivated. Our interpretation of QFI is based on the hierarchy and redistribution of spectral weight in $S(\mathbf{k}, \omega; T)$ across spinon- and photon-dominated frequency windows. In particular, QFI is a frequency integral of the dynamical response with a smooth, temperature-dependent kernel, so the qualitative crossover structure is controlled primarily by where spectral weight resides relative to the relevant excitation scales, rather than by precise ground-state convergence or microscopic level spacings at finite N . We therefore use the $N = 16$ cluster as the largest symmetry-respecting finite- T reference and benchmark the resulting spectral hierarchy and QFI trends against complementary thermodynamic-limit approaches discussed in the main text.

Separately, typicality fluctuations can increase at low temperature and must be controlled independently of physical finite-size effects. We therefore average over 64 random-vector realizations and quantify the resulting spread. To this end, we have performed rigorous convergence as shown in Fig. S9. The disorder-averaged QFI converges as $\sigma/\sqrt{n}$, and with $n = 64$ realizations the standard error of the mean remains below 1% of the signal at low temperatures and below $\sim 5\%$ even at the worst-case temperature.

Besides the results shown in the main text, we also include the $S^{\pm}$ channel at the X point as well as the S^z channel in Fig. S10. We can also clearly observe the delineation of the two temperature scales from the computed QFI values, as we discussed in the main text. Furthermore, we would like to highlight the increasing magnitude $f_Q(S_{\mathbf{q}}^z)$ as we move towards the Heisenberg point $J_{\pm} = -0.5$. QFI also encapsulates the total fluctuation generated by $e^{i\theta O}$ via some operator O . In the $U(1)$ lattice gauge theory description, $S^z \sim E$, the canonical electric field. Therefore, at the lowest temperature, $f_Q(S_{\mathbf{q}}^z) \sim \langle EE \rangle$ can be thought of as some measure of the total

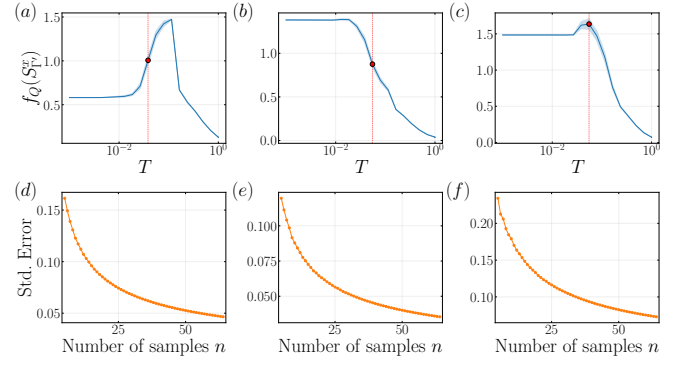


FIG. S9. Quantum Fisher information density $f_Q(S_{\Gamma}^x)$ for $J_{\pm} = 0.05$ (a), $J_{\pm} = -0.15$ (b), $J_{\pm} = -0.3$ (c) as a function of time averaged over 64 samples. The shaded blue region denotes the standard error, and the red dots highlight the temperature associated with the largest error. We then perform their respective convergence analysis at the temperature with the largest error for $J_{\pm} = 0.05, -0.15, -0.3$ respectively in (d,e,f).

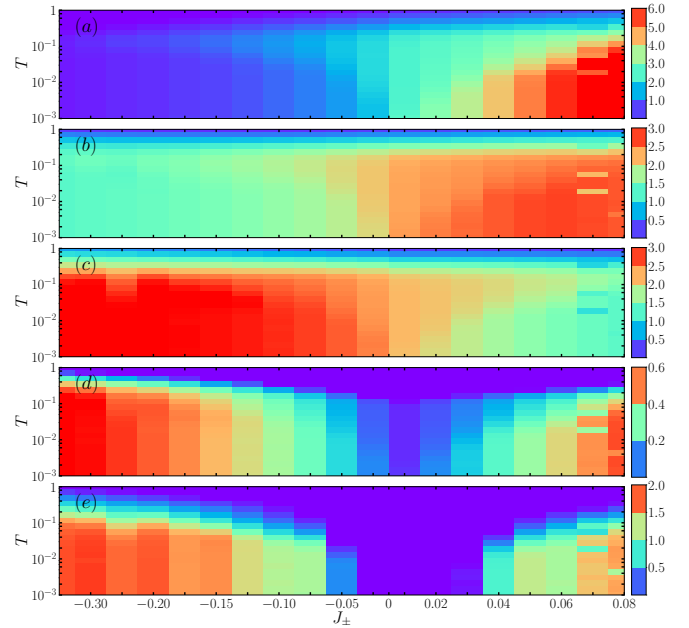


FIG. S10. The ED calculation results of the QFI density f_Q (a-c) in $S^{\pm}$ channel and (d-e) in S^z channel as a function of temperature T and $J_{\pm}$. (a) illustrates f_Q at $\Gamma = (0, 0, 0)$ while (b) is that at X and (c) at $\Gamma' = (4\pi, 4\pi, 0)$. (d) tells f_Q at X while (e) is that at $\Gamma' = (4\pi, 4\pi, 0)$.

gauge fluctuation.

III. GAUGE MEAN FIELD THEORY

We now provide a detailed account of the gauge mean-field theory (GMFT) formalism, which offers a canonical mapping of the spin Hamiltonian to that of a lattice $U(1)$ gauge theory coupled to some bosonic matter field. To keep the presentation transparent, let us first focus on the regime where the coupling

J_z is the dominant interaction, i.e. $|J_z| > |J_{xx}| = |J_{yy}|$, so that we may set $J_z = J_z$.

In this formulation, one introduces a slave “charge” degree of freedom on the sites of the parent diamond lattice, defined as

$$Q_{\mathbf{r}_\alpha} = \sum_{\mu \in \partial t_{\mathbf{r}_\alpha}} S_{\mathbf{R}_\mu}^z, \quad (\text{S18})$$

where $t_{\mathbf{r}_\alpha}$ denotes the tetrahedron centered at $\mathbf{r}_\alpha$, and $\partial t_{\mathbf{r}_\alpha}$ are the four pyrochlore spins forming its, μ denotes the sublattice indices. The index $\alpha \in A, B$ distinguishes the two diamond sublattices (up- and down-pointing tetrahedra). Notice we make an explicit distinction between the sublattice-indexed spin coordinates, denoted by $\mathbf{R}_\mu$, where $\mu \in \{0, 1, 2, 3\}$ represents the sublattice index, and the parent diamond lattice, denoted by the lower case $\mathbf{r}_\alpha$. These two coordinate systems are related by:

$$\mathbf{R}_\mu = \mathbf{r}_\alpha + \eta_\alpha \mathbf{b}_\mu / 2 \quad (\text{S19})$$

where $\mathbf{b}_\mu$ connects A-sublattice sites to their four nearest B-sublattice neighbors and $\eta_A = 1$ and $\eta_B = -1$.

$$\mathbf{b}_0 = -\frac{1}{4}(1, 1, 1) \quad (\text{S20a})$$

$$\mathbf{b}_1 = \frac{1}{4}(-1, 1, 1) \quad (\text{S20b})$$

$$\mathbf{b}_2 = \frac{1}{4}(1, -1, 1) \quad (\text{S20c})$$

$$\mathbf{b}_3 = \frac{1}{4}(1, 1, -1). \quad (\text{S20d})$$

By canonical construction, the conjugate variable $\varphi_{\mathbf{r}_\alpha}$ obeys $[\varphi_{\mathbf{r}_\alpha}, Q_{\mathbf{r}_\alpha'}] = i\delta_{\mathbf{r}_\alpha, \mathbf{r}_\alpha'}$, allowing one to define bosonic (spinon) raising and lowering operators

$$\Phi_{\mathbf{r}_\alpha}^\dagger = e^{i\varphi_{\mathbf{r}_\alpha}}, \quad \Phi_{\mathbf{r}_\alpha} = e^{-i\varphi_{\mathbf{r}_\alpha}}. \quad (\text{S21})$$

The original pseudospin operators can now be re-expressed in the enlarged Hilbert space $\mathcal{H} = \mathcal{H}_Q \otimes \mathcal{H}_{\text{spin}}$, where the mapping is given by

$$S_{\mathbf{R}_\mu}^+ \rightarrow \Phi_{\mathbf{r}_A}^\dagger \left(\frac{1}{2} e^{iA_{\mathbf{r}_A, \mathbf{r}_A} + \mathbf{b}_\mu} \right) \Phi_{\mathbf{r}_A + \mathbf{b}_\mu}, \quad (\text{S22})$$

$$S_{\mathbf{R}_\mu}^z \rightarrow E_{\mathbf{r}_A, \mathbf{r}_A + \mathbf{b}_\mu}, \quad (\text{S23})$$

with $A_{\mathbf{r}_A, \mathbf{r}_A + \mathbf{b}_\mu}$ and $E_{\mathbf{r}_A, \mathbf{r}_A + \mathbf{b}_\mu}$ denoting conjugate gauge and electric fields on the diamond links.

Physically, this construction makes explicit the emergent gauge structure inherent in the spin-ice manifold: spin flips map to spinon matter hopping minimally coupled to compact $U(1)$ gauge fields.

The resulting Hamiltonian contains quadratic spinon charge terms and spinon hopping terms ($J_\pm$), yielding

$$\mathcal{H} = \frac{J_z}{2} \sum_{\mathbf{r}_\alpha} Q_{\mathbf{r}_\alpha}^2 - \frac{J_\pm}{4} \sum_{\mathbf{r}_\alpha} \sum_{\mu, \nu \neq \mu} \Phi_{\mathbf{r}_\alpha + \eta_\alpha \mathbf{b}_\mu}^\dagger \Phi_{\mathbf{r}_\alpha + \eta_\alpha \mathbf{b}_\nu}$$

$$e^{i\eta_\alpha (A_{\mathbf{r}_\alpha, \mathbf{r}_\alpha + \eta_\alpha \mathbf{b}_\nu} - A_{\mathbf{r}_\alpha, \mathbf{r}_\alpha + \eta_\alpha \mathbf{b}_\mu})}, \quad (\text{S24})$$

where $J_\pm = -(J_{xx} + J_{yy})/4$.

At this stage, we adopt two crucial approximations: (i) the electric field E is integrated out, leaving behind purely bosonic matter fields coupled to static background fluxes; and (ii) the gauge field A is frozen to its mean-field value $\bar{A}$, thereby neglecting dynamical gauge fluctuations. These simplifications transform the model into a tractable quadratic bosonic theory, whose self-consistent solution captures the stability of the $U(1)$ quantum spin liquid. Evaluating the Hamiltonian at the spin ice manifold where $Q = 0$, we obtained that:

$$\mathcal{H} = -\frac{J_\pm}{4} \sum_{\mathbf{r}_\alpha} \sum_{\mu, \nu \neq \mu} \Phi_{\mathbf{r}_\alpha + \eta_\alpha \mathbf{b}_\mu}^\dagger \Phi_{\mathbf{r}_\alpha + \eta_\alpha \mathbf{b}_\nu} e^{i\eta_\alpha (\bar{A}_{\mathbf{r}_\alpha, \mathbf{r}_\alpha + \eta_\alpha \mathbf{b}_\nu} - \bar{A}_{\mathbf{r}_\alpha, \mathbf{r}_\alpha + \eta_\alpha \mathbf{b}_\mu})}. \quad (\text{S25})$$

To compute QFI, we need to calculate the dynamical spin susceptibility. Under the GMFT formalism, since the photonic mode is integrated out, we only have access to the spinon dynamics. In other words, we only have access to $\langle S^+ S^- \rangle \sim \langle \Phi^\dagger \Phi \Phi^\dagger \Phi \rangle$. In particular, we can compute the spin structure factor via the following formalism:

$$\begin{aligned} S_{\mu\nu}^{+-}(\mathbf{q}, \omega) &= \frac{1}{N} \sum_{\mathbf{R}_\mu, \mathbf{R}_\nu'} e^{i\mathbf{q} \cdot (\mathbf{R}_\mu - \mathbf{R}_\nu')} \int dt e^{i\omega t} \langle S_{\mathbf{R}_\mu}^+(t) S_{\mathbf{R}_\nu'}^-(0) \rangle \\ &= \frac{1}{N} \sum_{\mathbf{r}_A, \mathbf{r}_A'} \sum_{\mu, \nu} e^{i\mathbf{q} \cdot (\mathbf{r}_A - \mathbf{r}_A' + (\mathbf{b}_\mu - \mathbf{b}_\nu)/2)} \\ &\int dt e^{i\omega t} \frac{1}{4} \langle \Phi_{\mathbf{r}_A}^\dagger(t) e^{i\bar{A}_{\mathbf{r}_A, \mathbf{r}_A} + \mathbf{b}_\mu} \Phi_{\mathbf{r}_A + \mathbf{b}_\mu}(t) \Phi_{\mathbf{r}_A' + \mathbf{b}_\nu}^\dagger(0) e^{-i\bar{A}_{\mathbf{r}_A', \mathbf{r}_A'} + \mathbf{b}_\nu} \Phi_{\mathbf{r}_A'}(0) \rangle \end{aligned} \quad (\text{S26})$$

where N is the number of spins. We can then evaluate this using Wick's contraction. Within the XXZ model, it turns out that the only non-trivial term is:

$$\begin{aligned} S_{\mu\nu}^{+-}(\mathbf{q}, \omega) &= \frac{1}{N} \sum_{\mathbf{r}_A, \mathbf{r}_A'} \sum_{\mu, \nu} e^{i\mathbf{q} \cdot (\mathbf{r}_A - \mathbf{r}_A' + (\mathbf{b}_\mu - \mathbf{b}_\nu)/2)} \\ &\int dt e^{i\omega t} \frac{F_{\mu\nu}}{4} \langle \Phi_{\mathbf{r}_A + \mathbf{b}_\mu}(t) \Phi_{\mathbf{r}_A' + \mathbf{b}_\nu}^\dagger(0) \rangle \langle \Phi_{\mathbf{r}_A}^\dagger(t) \Phi_{\mathbf{r}_A'}(0) \rangle, \end{aligned} \quad (\text{S27})$$

where $F_{\mu\nu} = e^{i(\bar{A}_{\mathbf{r}_A, \mathbf{r}_A} + \mathbf{b}_\nu - \bar{A}_{\mathbf{r}_A, \mathbf{r}_A} + \mathbf{b}_\mu)}$. In other words, we can compute the dynamical spin susceptibility by evaluating the Green's function of the spinons. We can then integrate this to find the QFI of the spinon channel at zero temperature.

We show the QFI computed from GMFT at zero temperature in Fig. S11. Despite having made drastic assumptions in the theory, the results of GMFT actually align quite well with the QMC results (up to a factor of approximately 6/7).

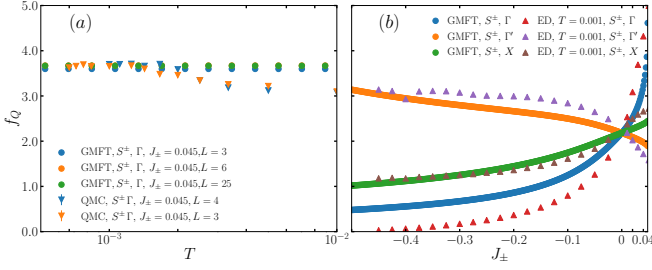


FIG. S11. The GMFT calculation results of the QFI density f_Q in the $S^\pm$ channel. Panel (a) shows the comparison between GMFT results (with a factor 6/7) for various system sizes from $L = 3$ to $L = 25$ at $J_\pm = 0.045$, and the QMC results for $L = 3$ and $L = 4$. Panel (b) shows the QFI density as a function of $J_\pm$ at zero temperature in the $S^\pm$ channel at Γ , Γ' , and X .

IV. NEUTRON SCATTERING CROSS SECTION OF DIPOLAR-OCTUPOLAR MATERIAL CANDIDATES

The neutron scattering cross section is obtained via Fermi's golden rule, and after tracing over all the sample spin degrees of freedom, we obtained that:

$$\frac{d^2\sigma}{d\Omega d\omega} \propto \frac{k_f}{k_i} \sum_{\alpha\beta} (\delta_{\alpha\beta} - \hat{\mathbf{q}}_\alpha \cdot \hat{\mathbf{q}}_\beta) M^{\alpha\beta}(\mathbf{q}, \omega), \quad (\text{S28})$$

where k_f (k_i) is the norm of the momentum for the outgoing (incoming) neutron (in what follows, we will drop the kinetic prefactor by assuming $k_f/k_i \approx 1$ for the experiment of interest), and

$$M^{\alpha\beta}(\mathbf{q}, \omega) = \frac{1}{2\pi} \int dt \langle M^\alpha(-\mathbf{q}, t) M^\beta(\mathbf{q}, 0) \rangle e^{i\omega t}. \quad (\text{S29})$$

Here $M^\alpha(\mathbf{q}, \omega)$ is the Fourier transformed magnetization density operator.

In the case of DO pyrochlore compounds, the lowest lying crystal electric field doublet $|\pm\rangle$ can be modeled using pseudospin operators $\tau^z = \frac{1}{2}(|+\rangle\langle+| - |-\rangle\langle-|)$ and $\tau^\pm = |\pm\rangle\langle\mp|$ that transform non-trivially under point group operation [75, 130]. In the case of Cerium-based candidates, only the pseudospin component τ^z has a dipolar magnetic charge density and linearly couples with the magnetic field (e.g., $|\pm\rangle = |J = 7/2, m_J = \pm 3/2\rangle$ for $\text{Ce}_2\text{Zr}_2\text{O}_7$ [77, 96, 129]). The pseudospin components τ^z are defined with respect to the local z -axis, as determined by crystal electric field analysis. These local z -axes vary according to the tetrahedron sublattice index and are given by

$$\mathbf{z}_0 = \frac{1}{\sqrt{3}}(1, 1, 1) \quad (\text{S30a})$$

$$\mathbf{z}_1 = \frac{1}{\sqrt{3}}(1, -1, -1) \quad (\text{S30b})$$

$$\mathbf{z}_2 = \frac{1}{\sqrt{3}}(-1, 1, -1) \quad (\text{S30c})$$

$$\mathbf{z}_3 = \frac{1}{\sqrt{3}}(-1, -1, 1). \quad (\text{S30d})$$

Therefore, in the dipolar approximation, the magnetization density operator is:

$$\begin{aligned} M^\alpha(\mathbf{q}, t) &\propto \frac{1}{\sqrt{N}} \sum_{\mathbf{R}_\mu} e^{i\mathbf{q}\cdot\mathbf{R}_\mu} \hat{\mathbf{z}}_\mu^\alpha \tau_{\mathbf{R}_\mu}^z(t) \\ &= \sum_{\mu} \hat{\mathbf{z}}_\mu^\alpha \tau_{\mathbf{q},\mu}^z(t), \end{aligned} \quad (\text{S31})$$

where we have defined $\tau_{\mathbf{q},\mu}^z(t) = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}_\mu} e^{i\mathbf{q}\cdot\mathbf{R}_\mu} \tau_{\mathbf{R}_\mu}^z(t)$ via the standard Fourier transform.

Plugging this into Eq. (S28) and (S29):

$$\begin{aligned} \frac{d^2\sigma}{d\Omega d\omega} &\propto \frac{1}{N} \sum_{\alpha\beta\mu\nu} (\delta_{\alpha\beta} - \hat{\mathbf{q}}_\alpha \cdot \hat{\mathbf{q}}_\beta) (\hat{\mathbf{z}}_\mu^\alpha (\hat{\mathbf{z}}_\nu^\beta) \langle \tau_{-\mathbf{q},\mu}^z(t) \tau_{\mathbf{q},\nu}^z(0) \rangle) \\ &= \frac{1}{N} \sum_{\mu\nu} \left(\hat{\mathbf{z}}_\mu \cdot \hat{\mathbf{z}}_\nu - \frac{(\hat{\mathbf{z}}_\mu \cdot \mathbf{Q})(\hat{\mathbf{z}}_\nu \cdot \mathbf{Q})}{|\mathbf{Q}|^2} \right) \langle \tau_{-\mathbf{q},\mu}^z(t) \tau_{\mathbf{q},\nu}^z(0) \rangle \\ &:= A^{\text{DO}}(\mathbf{q}, t). \end{aligned} \quad (\text{S32})$$

$$:= A^{\text{DO}}(\mathbf{q}, t). \quad (\text{S33})$$

We can then write out explicitly the transverse projector in a DO pyrochlore system for the $\langle \tau^z \tau^z \rangle$ channel as:

$$\mathcal{F}_{\mu\nu} := \hat{\mathbf{z}}_\mu \cdot \hat{\mathbf{z}}_\nu - \frac{(\hat{\mathbf{z}}_\mu \cdot \mathbf{Q})(\hat{\mathbf{z}}_\nu \cdot \mathbf{Q})}{|\mathbf{Q}|^2} \quad (\text{S34})$$

In this work, we want to compute experimentally relevant QFI for DO compounds, which are generally described by an XYZ Hamiltonian with a finite $\tilde{J}_{xz}$:

$$\mathcal{H}_{\text{DO}} = \sum_{\langle i,j \rangle} T_{\alpha\alpha} \tau_i^\alpha \tau_j^\alpha + T_{xz} (\tau_i^x \tau_j^z + \tau_i^z \tau_j^x) - g\mu_B \sum_i \mathbf{B}_i \cdot \hat{\mathbf{z}}_i, \quad (\text{S35})$$

where $\alpha \in \{x, y, z\}$. We can remove the T_{xz} term by applying a rotation along the y -axis: $\tau^y = \tilde{S}^y$; $\tau^x = \cos(\theta)\tilde{S}^x - \sin(\theta)\tilde{S}^z$; $\tau^z = \sin(\theta)\tilde{S}^x + \cos(\theta)\tilde{S}^z$; $\tan(2\theta) = \frac{2T_{xz}}{T_{xx} - T_{zz}}$. By doing so, we map this problem to the XYZ model:

$$\mathcal{H}_{\text{XYZ}} = \sum_{\langle i,j \rangle} \tilde{J}_{\alpha\alpha} \tilde{S}_i^\alpha \tilde{S}_j^\alpha - g\mu_B \sum_i \mathbf{B}_i \cdot (\tilde{S}_i^z \cos\theta + \tilde{S}_i^x \sin\theta). \quad (\text{S36})$$

Finally, we cover the XXZ model described in Eq. (1) of the main text when under the simplifying assumption that $\tilde{J}_{\alpha\alpha} = \tilde{J}_{\beta\beta} = -2J_\pm < \tilde{J}_{\gamma\gamma} = J_z$ with one dominant component, where α, β, γ are the 3 pseudospin components. In doing so, we map $S^z = \tilde{S}^\gamma$ and the other two components to be x and y in Eq. (1) of the main text. Let us define

$$C_{\mu\nu}^{\alpha\beta}(\mathbf{q}, \omega) = \frac{1}{2\pi} \int dt e^{i\omega t} \langle S_{-\mathbf{q},\mu}^\alpha(t) S_{\mathbf{q},\nu}^\beta(0) \rangle \quad (\text{S37})$$

In the case of $\text{Ce}_2\text{Zr}_2\text{O}_7$, the proposed parameter sets are (0.062, 0.063, 0.011) meV and (0.063, 0.062, 0.011) meV with a potentially small $\theta \lesssim 0.1\pi$. These two sets have equally excellent goodness of fit [87]. As such, depending on whether if J_{xx} or J_{yy} is dominant, $\tilde{S}^z = (S^+ - S^-)/2i$ or $\tilde{S}^z = (S^+ + S^-)/2$

as either $\tilde{S}^x$ or $\tilde{S}^y$ becomes the dominant longitudinal component. Therefore, in the case of $\text{Ce}_2\text{Zr}_2\text{O}_7$:

$$A^{\text{DO}}(\mathbf{q}, \omega) \sim \sum_{\mu, \nu} \frac{\mathcal{F}_{\mu\nu}}{4} [(C_{\mu\nu}^{+-}(\mathbf{q}, \omega) + C_{\mu\nu}^{-+}(\mathbf{q}, \omega)) \pm (C_{\mu\nu}^{++}(\mathbf{q}, \omega) + C_{\mu\nu}^{--}(\mathbf{q}, \omega))], \quad (\text{S38})$$

where the $\pm$ sign depends on whether if $\tilde{S}^x$ or $\tilde{S}^y$ is dominant. Under the XXZ model, the latter two terms vanish as a consequence of the $U(1)$ symmetry of the Hamiltonian. As such, to compute A^{DO} :

$$A^{\text{DO}}(\mathbf{q}, \omega) = \sum_{\mu, \nu} \mathcal{F}_{\mu\nu} (C_{\mu\nu}^{+-}(\mathbf{q}, \omega) + C_{\mu\nu}^{-+}(\mathbf{q}, \omega)) / 4. \quad (\text{S39})$$

V. QUANTUM FISHER INFORMATION MATRIX

A. Review of the Quantum Fisher Information

For this appendix to be as self-contained as possible, we will first review some basic facts about the QFI before discussing more specifically its relation to spin structure factors and neutron scattering cross sections.

For a differentiable family of quantum states $\rho_\theta = e^{-i\theta O} \rho e^{i\theta O}$ generated by a Hermitian operator O , the quantum Fisher information (QFI) is defined via the symmetric logarithmic derivative (SLD) L as [114]

$$F_Q[\rho, O] \equiv \text{Tr}(\rho L^2), \quad \partial_\theta \rho|_{\theta=0} = \frac{1}{2} \{\rho, L\} = -i[\rho, O].$$

In eigenbasis $\rho = \sum_n p_n |n\rangle\langle n|$, the QFI admits the Lehmann spectral representation

$$F_Q[\rho, O] = 2 \sum_{m, n} \frac{(p_m - p_n)^2}{p_m + p_n} |\langle m|O|n\rangle|^2,$$

where terms with $p_m + p_n = 0$ are omitted. For pure states $\rho = |\psi\rangle\langle\psi|$, this reduces to

$$F_Q[|\psi\rangle, O] = 4(\langle\psi|O^2|\psi\rangle - \langle\psi|O|\psi\rangle^2) = 4\text{Var}_\psi(O).$$

For thermal states $\rho \propto e^{-\beta H}$, the same expression holds with $p_n = e^{-\beta E_n} / Z$ and the $\{|n\rangle\}$ chosen as the energy eigenbasis of H .

A pure N -particle state is called m -producible if it factorizes into a tensor product of blocks, each containing at most m parties; a mixed state is m -producible if it is a convex mixture of such pure states. The entanglement depth is the smallest m for which the state is m -producible. For a generator that is a sum of local terms,

$$O = \sum_{i=1}^N O_i, \quad \Delta_i \equiv \lambda_{\max}(O_i) - \lambda_{\min}(O_i),$$

where Δ_i is the spectral width of O_i and $\lambda_{\max(\min)}$ are the corresponding largest (smallest) eigenvalues, every m -producible

state ρ satisfies the QFI bound (Hyllus–Tóth) [35, 36]

$$F_Q[\rho, O] \leq \max_{\substack{\text{partitions } \{B_\ell\} \\ |B_\ell| \leq m}} \sum_{\ell} \left(\sum_{i \in B_\ell} \Delta_i \right)^2, \quad (\text{S40})$$

where the maximum runs over all partitions of $\{1, \dots, N\}$ into blocks B_ℓ with $|B_\ell| \leq m$.

If $\Delta_i = \Delta$ for all i , then (S40) becomes

$$F_Q[\rho, O] \leq \Delta^2 \sum_{\ell} |B_\ell|^2.$$

Thus, for fixed $N = \sum_{\ell} |B_\ell|$ and the constraint $|B_\ell| \leq m$, maximizing the bound reduces to maximizing $\sum_{\ell} |B_\ell|^2$. The optimal partition is therefore as many size- m blocks as possible and at most one residual block. Writing $N = sm + r$ with $s = \lfloor N/m \rfloor$ and $0 \leq r < m$, the maximizing partition has s blocks of size m and, if $r > 0$, one block of size r (all other blocks empty). Therefore

$$F_Q[\rho, O] \leq \Delta^2 (sm^2 + r^2), \quad (\text{S41})$$

with equality attained by that partition (if $r = 0$, it is s blocks of size m).

Hence, if a measured (or computed) QFI violates (S41) for some m , i.e. $F_Q[\rho, O] > \Delta^2 (sm^2 + r^2)$, then the state has entanglement depth at least $m+1$.

Remark 1 (inhomogeneous local spectra; finite- N bound). Let $O = \sum_{i=1}^N c_i O_i$ with $c_i \in \mathbb{R}$ and local spectral widths $\Delta_i := \lambda_{\max}(O_i) - \lambda_{\min}(O_i)$, so that the spectral width of $c_i O_i$ is $|c_i| \Delta_i$. Define the nonnegative weights

$$w_i := |c_i| \Delta_i.$$

For an m -producible state ρ , the Hyllus–Tóth bound (S40) implies the tight finite- N constraint

$$F_Q[\rho, O] \leq \max_{\substack{\text{partitions } \{B_\ell\} \\ |B_\ell| \leq m}} \sum_{\ell} \left(\sum_{i \in B_\ell} w_i \right)^2. \quad (\text{S42})$$

Moreover, since x^2 is convex, the maximizing partition in (S42) is obtained by grouping the largest weights together. Concretely, let $w_{(1)} \geq w_{(2)} \geq \dots \geq w_{(N)}$ be the nonincreasing rearrangement of $\{w_i\}$, and write $N = sm + r$ with $s = \lfloor N/m \rfloor$ and $0 \leq r < m$. Then the tight bound can be written explicitly as

$$F_Q[\rho, O] \leq \sum_{\ell=1}^s \left(\sum_{j=(\ell-1)m+1}^{\ell m} w_{(j)} \right)^2 + \left(\sum_{j=sm+1}^N w_{(j)} \right)^2, \quad (\text{S43})$$

with the last term absent if $r = 0$.

It is sometimes convenient to use a simpler, fully analytic (but generally non-tight) relaxation of (S42). Applying Cauchy–Schwarz on each block B_ℓ gives

$$\left(\sum_{i \in B_\ell} w_i \right)^2 \leq |B_\ell| \sum_{i \in B_\ell} w_i^2. \quad (\text{S44})$$

Summing (S44) over all blocks in the partition (including the residual block of size r if $N = sm + r$ with $0 \leq r < m$) yields

$$\sum_{\ell} \left(\sum_{i \in B_{\ell}} w_i \right)^2 \leq \sum_{\ell} |B_{\ell}| \sum_{i \in B_{\ell}} w_i^2. \quad (\text{S45})$$

Since every admissible block satisfies $|B_{\ell}| \leq m$ (and in particular the residual block has size $r < m$), each term on the right-hand side obeys

$$|B_{\ell}| \sum_{i \in B_{\ell}} w_i^2 \leq m \sum_{i \in B_{\ell}} w_i^2,$$

so that

$$\sum_{\ell} \left(\sum_{i \in B_{\ell}} w_i \right)^2 \leq m \sum_{\ell} \sum_{i \in B_{\ell}} w_i^2 = m \sum_{i=1}^N w_i^2. \quad (\text{S46})$$

Because (S46) holds for every admissible partition, it also holds after maximizing over partitions in (S42). We therefore obtain the relaxed, finite- N bound

$$F_Q[\rho, O] \leq m \sum_{i=1}^N w_i^2 = m \sum_{i=1}^N c_i^2 \Delta_i^2, \quad (\text{S47})$$

which is valid for arbitrary N and does not require m to divide N . We stress that the use of the factor m is a uniform upper bound that also covers the residual block (since $r < m$); keeping $|B_{\ell}|$ explicitly in (S45) yields a slightly tighter but partition-dependent expression.

Remark 2 (sum of witnesses and spectral-width control). Suppose we define a composite quantity as the sum of two QFI densities, $f_Q^{(\Sigma)} \equiv f_Q(O^{(1)}) + f_Q(O^{(2)})$, with additive generators $O^{(k)} = \sum_{i=1}^N O_i^{(k)}$, $\Delta_i^{(k)} := \lambda_{\max}(O_i^{(k)}) - \lambda_{\min}(O_i^{(k)})$. For any m -producible state ρ one has, by the standard block-partition argument and Cauchy–Schwarz,

$$f_Q(O^{(k)}) \leq m \overline{(\Delta^{(k)})^2}, \quad \overline{(\Delta^{(k)})^2} := \frac{1}{N} \sum_{i=1}^N (\Delta_i^{(k)})^2, \quad (\text{S48})$$

hence the sum bound

$$f_Q^{(\Sigma)} \leq m \left(\overline{(\Delta^{(1)})^2} + \overline{(\Delta^{(2)})^2} \right). \quad (\text{S49})$$

Therefore, the normalized sum witness

$$\frac{f_Q^{(\Sigma)}}{\overline{(\Delta^{(1)})^2} + \overline{(\Delta^{(2)})^2}} > m \implies \text{entanglement depth} \geq m + 1. \quad (\text{S50})$$

Equation (S49) is likewise a convenient sufficient (generally non-tight) bound obtained from Cauchy–Schwarz and block-size control; the corresponding tight finite- N bound follows from applying (S40) separately to each generator and maximizing over partitions.

B. Spin Structure Factors

In the main text, we have used $S_{\mathbf{q}}^{\alpha} = \sum_i S_i^{\alpha} e^{i\mathbf{q} \cdot \mathbf{R}_i}$ as the operator associated with the QFI. One would note that the usual definition of QFI requires an operator O to be Hermitian such that the induced generator $e^{i\theta O}$ is unitary and thereby generates norm-preserving dynamics. Although $S_{\mathbf{q}}^{\alpha}$ is not Hermitian, we show below that the definition of QFI in Eq. (4) of the main text still encompasses all the important properties of the conventional QFI. Namely, it is a robust measure of fluctuations, and we can still derive a meaningful lower bound of entanglement depth.

To keep our results general, let us consider some Hermitian operator $O_{\mathbf{R}_{\mu}}$ on each site under the coordinate system specified in Eq. (S19) such that the operator we associate with QFI has the form $O_{\mathbf{q}} = \sum_{\mathbf{R}_{\mu}} O_{\mathbf{R}_{\mu}} e^{i\mathbf{q} \cdot \mathbf{R}_{\mu}}$, where $O_{\mathbf{R}_{\mu}}$ is some local observable with homogeneous spectral width $\Delta_{\mathbf{R}_{\mu}} = \Delta$. For later discussion, let us denote such a homogeneous spectral width as $\Delta(O_{\mathbf{R}_{\mu}})$ to be explicit about the underlying local operators. We can plug in $O_{\mathbf{R}_{\mu}} = S_{\mathbf{R}_{\mu}}^{\alpha}$ where $\alpha \in \{x, y, z\}$ at the end to make a direct connection with the definitions used in the main text. To begin, let us split $O_{\mathbf{q}} = O_{c;\mathbf{q}} + iO_{s;\mathbf{q}}$ into an Hermitian and antihermitian parts, where $O_{c;\mathbf{q}}$ and $O_{s;\mathbf{q}}$ are Hermitian operators:

$$O_{c;\mathbf{q}} = \frac{O_{\mathbf{q}} + O_{\mathbf{q}}^{\dagger}}{2} = \sum_{\mathbf{R}_{\mu}} O_{\mathbf{R}_{\mu}} \cos(\mathbf{q} \cdot \mathbf{R}_{\mu}) \quad (\text{S51})$$

$$O_{s;\mathbf{q}} = \frac{O_{\mathbf{q}} - O_{\mathbf{q}}^{\dagger}}{2i} = \sum_{\mathbf{R}_{\mu}} O_{\mathbf{R}_{\mu}} \sin(\mathbf{q} \cdot \mathbf{R}_{\mu}) \quad (\text{S52})$$

Now we construct a QFI matrix (QFIM) F [133–135], whose elements are specified by:

$$F_{ab}(T) := 4 \int d\omega \tanh\left(\frac{\omega}{2T}\right) (1 - e^{-\omega/T}) \mathcal{A}_{ab}(\mathbf{q}, \omega). \quad (\text{S53})$$

where

$$\mathcal{A}_{ab}(\mathbf{q}, \omega) = \frac{1}{2\pi N} \int dt e^{i\omega t} \langle O_{a,\mathbf{q}}(t) O_{b,\mathbf{q}}(0) \rangle. \quad (\text{S54})$$

with $a, b \in \{c, s\}$. Now we wish to show that $f_Q(O_{\mathbf{q}}, T)$ defined in the main text via Eq. (4) is related to the trace of this QFIM. Since $O_{\mathbf{q}} = O_{c;\mathbf{q}} + iO_{s;\mathbf{q}}$,

$$\begin{aligned} O_{\mathbf{q}}^{\dagger} O_{\mathbf{q}} &= O_{c;\mathbf{q}} O_{c;\mathbf{q}} + O_{s;\mathbf{q}} O_{s;\mathbf{q}} + i[O_{c;\mathbf{q}}, O_{s;\mathbf{q}}] \\ &= O_{c;\mathbf{q}} O_{c;\mathbf{q}} + O_{s;\mathbf{q}} O_{s;\mathbf{q}}, \end{aligned} \quad (\text{S55})$$

as $[O_{c;\mathbf{q}}, O_{s;\mathbf{q}}] = 0$. Therefore, we see that

$$\begin{aligned} A(\mathbf{q}, \omega) &:= \frac{1}{2\pi N} \int dt \langle O_{\mathbf{q}}^{\dagger}(t) O_{\mathbf{q}}(0) \rangle e^{i\omega t} \\ &= \frac{1}{2\pi N} \int dt e^{i\omega t} \langle O_{c;\mathbf{q}}(t) O_{c;\mathbf{q}}(0) \rangle + \langle O_{s;\mathbf{q}}(t) O_{s;\mathbf{q}}(0) \rangle \\ &= \mathcal{A}_{cc} + \mathcal{A}_{ss}. \end{aligned} \quad (\text{S56})$$

As a result,

$$\begin{aligned} f_Q(O_q, T) &= \int d\omega \tanh\left(\frac{\omega}{2T}\right)(1 - e^{-\omega/T})A(\mathbf{q}, \omega) \\ &= \int d\omega \tanh\left(\frac{\omega}{2T}\right)(1 - e^{-\omega/T})(\mathcal{A}_{cc}(\mathbf{q}, \omega) + \mathcal{A}_{ss}(\mathbf{q}, \omega)) \\ &= f_Q(O_{c,q}, T) + f_Q(O_{s,q}, T) = \text{Tr}(F). \end{aligned} \quad (\text{S57})$$

In other words, the QFI used in the main text is precisely the trace of the QFI matrix introduced above. Interpreting F as a covariance matrix, this shows that the $f_Q(O_q, T)$ equals the total variance — the sum of the variances carried by all fluctuation modes.

Furthermore, $f_Q(O_q, T)$ provides a robust lower bound on the entanglement depth as $f_Q(O_q, T) = f_Q(O_{c,q}, T) + f_Q(O_{s,q}, T)$ is the well-bounded sum witness discussed in remark 2. Using the result outlined in the remark, for an m -producible state, we have

$$f_Q(O_q, T) \leq m \left((\overline{\Delta O_{c,q}})^2 + (\overline{\Delta O_{s,q}})^2 \right). \quad (\text{S58})$$

Finally, by negation,

$$f_Q(O_q, T) > m \left((\overline{\Delta O_{c,q}})^2 + (\overline{\Delta O_{s,q}})^2 \right) \quad (\text{S59})$$

implies the state is at least $(m + 1)$ -partite entangled.

As such, all that remains is to obtain a bound for these two channels. Using Eq. (S47) with inhomogeneous spectral width for each local operator $O_{\mathbf{R}_\mu}$, $\Delta_{\mathbf{R}_\mu} = \Delta(O_{\mathbf{R}_\mu})$, and $c_{\mathbf{R}_\mu} = \cos(\mathbf{q} \cdot \mathbf{R}_\mu)$,

$$\overline{(\Delta O_{c,q})^2} = \frac{(\Delta(O_{\mathbf{R}_\mu}))^2}{N} \sum_{\mathbf{R}_\mu} \cos^2(\mathbf{q} \cdot \mathbf{R}_\mu). \quad (\text{S60})$$

Therefore, by arguing the same for $O_{s,q}$, we can finally bound $f_Q(O_q)$ for any m -producible state:

$$\begin{aligned} f_Q(O_q) &\leq \frac{m(\Delta(O_{\mathbf{R}_\mu}))^2}{N} \sum_{\mathbf{R}_\mu} (\cos^2(\mathbf{q} \cdot \mathbf{R}_\mu) + \sin^2(\mathbf{q} \cdot \mathbf{R}_\mu)) \\ &= m(\Delta(O_{\mathbf{R}_\mu}))^2. \end{aligned} \quad (\text{S61})$$

As such, we find that $f_Q(O_q)$ is bounded above by $m(\Delta(O_{\mathbf{R}_\mu}))^2$. This is precisely the definition of the entanglement bound for conventional QFI [20, 45, 114] as if we have ignored the exponential part $e^{i\mathbf{q} \cdot \mathbf{R}_\mu}$ altogether.

To conclude, for an m -producible state, any operator of the form $O_q = \sum_{\mathbf{R}_\mu} O_{\mathbf{R}_\mu} e^{i\mathbf{q} \cdot \mathbf{R}_\mu}$, where $O_{\mathbf{R}_\mu}$ is some observable with homogeneous spectrum $\pm 1/2$ across all sites, has the corresponding QFI bound

$$f_Q(O_q, T) \leq m(\Delta(O_{\mathbf{R}_\mu}))^2. \quad (\text{S62})$$

Therefore, for $O_q = S_q^\alpha$ where $\alpha \in \{x, y, z\}$ in a spin-1/2 system:

$$f_Q(S_q^\alpha, T) > m(\Delta(S_{\mathbf{R}_\mu}^\alpha))^2 = m(1/2 - (-1/2))^2 = m, \quad (\text{S63})$$

implies that the state is at least $(m + 1)$ -partite entangled, regardless of momentum position $\mathbf{q}$.

Now, since we have established a well-constructed QFI for non-Hermitian operators, we can extend this even further for $\alpha = \pm$ in the main text. As discussed in the main text, $f_Q(S_q^\pm, T) = f_Q(S_q^x, T) + f_Q(S_q^y, T)$ is a sum witness. Therefore, by remark 2, for an m -producible state:

$$f_Q(S_q^\pm, T) = f_Q(S_q^x, T) + f_Q(S_q^y, T) \leq m(1 + 1) = 2m. \quad (\text{S64})$$

Equivalently, this is to say that if the spinon channel has nQFI

$$\text{nQFI}(S_q^\alpha) = f_Q(S_q^\pm, T)/2 > m, \quad (\text{S65})$$

then the state is at least $(m + 1)$ -partite entangled.

C. Neutron Scattering Cross Section

Similarly, for A^{DO} , to have a meaningful interpretation of the entanglement depth, we would have to determine $\lambda_{\max}$ and $\lambda_{\min}$ of the operator in which we are evaluating our QFI to determine the QFI spectral bound η . To do so, first, we should determine the exact operator O that the neutron scattering cross section contains. Let us define operators

$$S_q^{\text{NSF}} = \sum_{\mathbf{R}_\mu} (\hat{\mathbf{p}} \cdot \hat{\mathbf{z}}_\mu) \tau_{\mathbf{R}_\mu}^z e^{i\mathbf{q} \cdot \mathbf{R}_\mu} \quad (\text{S66a})$$

$$S_q^{\text{SF}} = \sum_{\mathbf{R}_\mu} (\hat{\mathbf{v}} \cdot \hat{\mathbf{z}}_\mu) \tau_{\mathbf{R}_\mu}^z e^{i\mathbf{q} \cdot \mathbf{R}_\mu}. \quad (\text{S66b})$$

Here $\hat{\mathbf{p}}$ denotes the neutron polarization, which is perpendicular to the momentum transfer ($\hat{\mathbf{v}} \cdot \hat{\mathbf{p}} = 0$), and $\hat{\mathbf{v}}$ is a vector perpendicular to both the polarization vector and the neutron momentum transfer ($\hat{\mathbf{v}} \cdot \hat{\mathbf{p}} = \hat{\mathbf{v}} \cdot \hat{\mathbf{q}} = 0$). These channels correspond to polarized neutron scattering experiments in the spin-flip/non-spin-flip (SF/NSF) channels with respect to the neutron polarization vector $\hat{\mathbf{p}}$. By definition,

$$A^{\text{DO}} \sim \langle S_q^{\text{NSF}\dagger} S_q^{\text{NSF}} \rangle + \langle S_q^{\text{SF}\dagger} S_q^{\text{SF}} \rangle, \quad (\text{S67})$$

where we sum up all transverse modes, which is equivalent to taking the transverse projector in Eq. (S32). Accordingly, we assemble a 2×2 QFIM F^{DO} from the two transverse components—taken here as the NSF and SF channels—so that the cross section A^{DO} defined in Eq. (S32) is identified with $\text{Tr } F^{\text{DO}}$. This construction is necessary because the transverse projector has two independent components. Hence, there is no single scalar operator S^{DO} with $A^{\text{DO}} \sim \langle S^{\text{DO}\dagger} S^{\text{DO}} \rangle$. Nevertheless, we still denote the QFI related to A^{DO} as $f_Q(S_q^{\text{DO}}, T)$ for consistency's sake. The operators entering F^{DO} are generally non-Hermitian, which is admissible since the bounds derived in the previous section apply directly. Applying Eq. (S62) to each channel then yields a lower bound on the entanglement depth. Finally, to bound the QFI induced by unpolarized neutron scattering, let us denote the corresponding QFI from the A^{DO} channel as the sum witness in remark 2 by

$$f_Q(S_q^{\text{DO}}) = f_Q(S_q^{\text{NSF}}) + f_Q(S_q^{\text{SF}}). \quad (\text{S68})$$

Therefore, let us bound the NSF and SF channels to bound the total unpolarized neutron scattering cross section. To do so, we have to determine λ_{max} and λ_{min} of each basis operator to get an actual interpretation of the entanglement depth. To do so, we need to evaluate at a specific momentum position $\mathbf{q}$. For clarity of later discussion, let us define vector Π_{NSF} and Π_{SF} with four components corresponding to the four pyrochlore sublattices μ as

$$\Pi_{NSF}^\mu(\mathbf{q}) = \hat{\mathbf{p}} \cdot \hat{\mathbf{z}}_\mu \quad (\text{S69a})$$

$$\Pi_{SF}^\mu(\mathbf{q}) = \hat{\mathbf{v}} \cdot \hat{\mathbf{z}}_\mu. \quad (\text{S69b})$$

Since, in principle, the operator $S_{\mathbf{q}}^{(N)SF}$ is non-Hermitian, we should break it down into the same cosine and sine channels again and discuss the sum of the spectral width on each channel. By Eq. (S58), we need to bound the cosine and the sine channel again. Let us denote $S_{c(s),\mathbf{q}}^{(N)SF}$ as the corresponding cosine and sine channels. Then, using Eq. (S47) with $c_\mu = \Pi_{(N)SF}^\mu$:

$$\overline{(\Delta S_{c,\mathbf{q}}^{(N)SF})^2} = \frac{1}{N} \sum_{\mathbf{R}_\mu} \cos^2(\mathbf{q} \cdot \mathbf{R}_\mu) |\Pi_{(N)SF}^\mu|^2. \quad (\text{S70})$$

Therefore, using Eq. (S58), we can derive a QFI bound on the NSF and SF channel separately for an m -producible state:

$$\begin{aligned} f_Q(S_{\mathbf{q}}^{(N)SF}, T) &\leq \frac{m}{N} \sum_{\mathbf{R}_\mu} (\cos^2(\mathbf{q} \cdot \mathbf{R}_\mu) + \sin^2(\mathbf{q} \cdot \mathbf{R}_\mu)) |\Pi_{(N)SF}^\mu|^2 \\ &= \frac{m}{N} \sum_{\mathbf{R}_\mu} |\Pi_{(N)SF}^\mu|^2 = \frac{m}{4} \sum_{\mu} |\Pi_{(N)SF}^\mu|^2. \end{aligned} \quad (\text{S71})$$

In the last line, we have used the fact that $\Pi_{(N)SF}^\mu$ depends only on the sublattice but not on the unit cell position. We can then sum over all unit cells, which gives a prefactor of $N/4$. Finally, the upper bound on $f_Q(S_{\mathbf{q}}^{DO})$ for an m -producible state is

$$f_Q(S_{\mathbf{q}}^{DO}) = f_Q(S_{\mathbf{q}}^{NSF}) + f_Q(S_{\mathbf{q}}^{SF}) \leq \frac{m}{4} \sum_{\mu} (|\Pi_{NSF}^\mu|^2 + |\Pi_{SF}^\mu|^2). \quad (\text{S72})$$

Now, to actually evaluate this bound, let us write

$$\sum_{\mu} |\Pi_{(N)SF}^\mu|^2 = \sum_{\mu} |\hat{\mathbf{a}} \cdot \hat{\mathbf{z}}_\mu|^2 = \hat{\mathbf{a}}^\top \left(\sum_{\mu} \hat{\mathbf{z}}_\mu \hat{\mathbf{z}}_\mu^\top \right) \hat{\mathbf{a}} = \hat{\mathbf{a}}^\top A \hat{\mathbf{a}}, \quad (\text{S73})$$

where $A := \sum_{\mu} \hat{\mathbf{z}}_\mu \hat{\mathbf{z}}_\mu^\top$ and $\hat{\mathbf{a}} = \hat{\mathbf{p}}$ if NSF and $\hat{\mathbf{a}} = \hat{\mathbf{v}}$ if SF.

For a single pyrochlore tetrahedron, take the four local $\hat{\mathbf{z}}_\mu$, a direct sum gives

$$A = \sum_{\mu=0}^3 \hat{\mathbf{z}}_\mu \hat{\mathbf{z}}_\mu^\top = \frac{4}{3} \mathbb{1}_{3 \times 3}, \quad (\text{S74})$$

since all off-diagonal entries cancel by symmetry (the signs appear equally often with opposite parity) and $\text{Tr} A = \sum_{\mu} \|\hat{\mathbf{z}}_\mu\|^2 = 4$ fixes the proportionality constant. Hence

$$\sum_{\mu} |\Pi_{(N)SF}^\mu|^2 = \frac{4}{3} \|\hat{\mathbf{a}}\|^2 = \frac{4}{3}, \quad (\text{S75})$$

which is independent of the direction of the unit vector $\hat{\mathbf{a}}$. As a result, for an m -producible state, QFI upper bounds on NSF, SF, and the total scattering channel are given by

$$f_Q(S_{\mathbf{q}}^{(N)SF}, T) \leq \frac{m}{4} \sum_{\mu} |\Pi_{(N)SF}^\mu|^2 = \frac{m}{3}. \quad (\text{S76a})$$

$$f_Q(S_{\mathbf{q}}^{DO}, T) \leq \frac{2m}{3}. \quad (\text{S76b})$$

To summarize, we derived a bound independent of $\mathbf{q}$ for the NSF, SF, and the unpolarized scattering channel with corresponding nQFI:

$$\text{nQFI}(S_{\mathbf{q}}^{(N)SF}, T) = 3 f_Q(S_{\mathbf{q}}^{(N)SF}, T) \quad (\text{S77a})$$

$$\text{nQFI}(S_{\mathbf{q}}^{DO}, T) = \frac{3}{2} f_Q(S_{\mathbf{q}}^{(N)SF}, T). \quad (\text{S77b})$$

We again stress that this bound is not the optimal bound in general but an upper bound of such via Cauchy-Schwartz as shown in Eq. (S71). The true optimal bound depends strongly on the choice of $\hat{\mathbf{p}}$ and $\hat{\mathbf{v}}$ and thereby the incident neutron momentum $\hat{\mathbf{q}}$. In theory, one would need to apply Eq. (S40) carefully to find the best bound.